

EE 6271- BASIC ELETRONICS

LECTURE 1b:

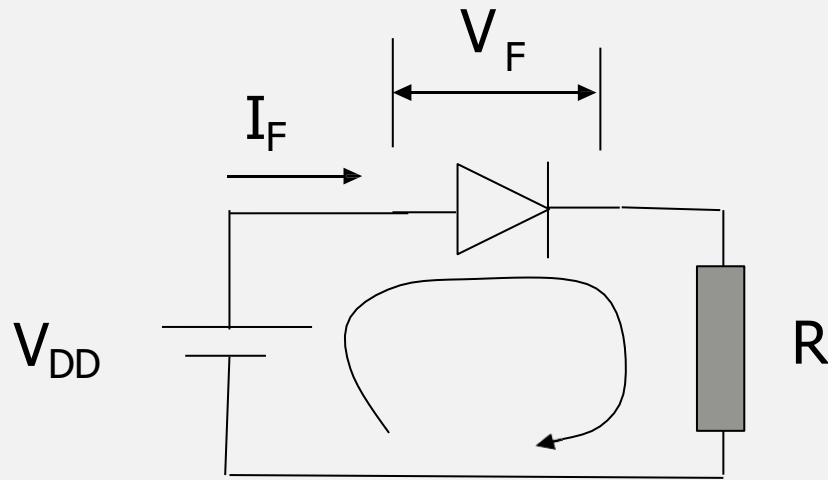
SEMICONDUCTOR DIODES AND APPLICATIONS

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Tuesday, April 21, 2026

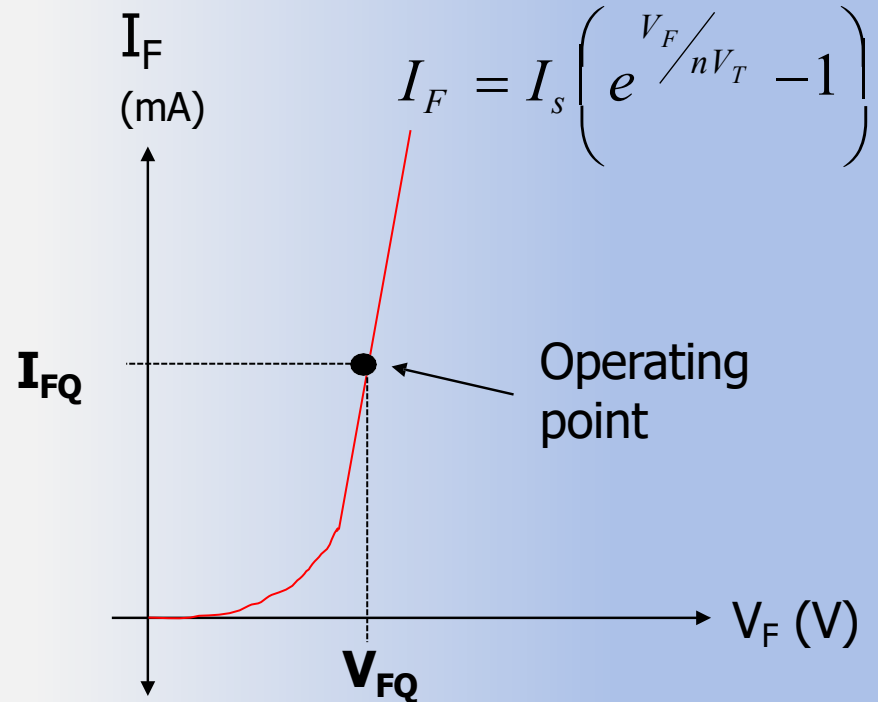
2025/2026

LOAD LINE AND OPERATING POINT(Q-POINT)



DC load line equation

$$V_{DD} = V_F + I_F R$$



Diode I-V characteristic

- ❑ The equation obtained from the diode circuit is called **Load-line Equation**.
- ❑ This equation gives us all possible current values for all the possible voltages values obtained across the diode in a given circuit.

OPERATING POINT...

- The values of V_F and I_F at operating point (V_{FQ} , I_{FQ}) must satisfy the two equations i.e.

- **Diode Equation**

- **DC load line equation**

- From the dc load line equation:

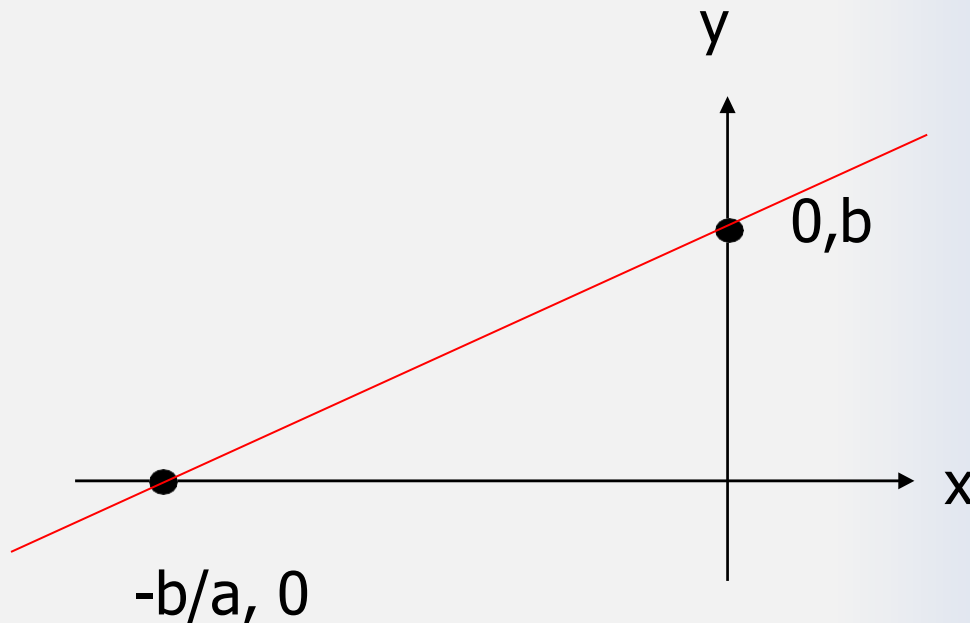
$$V_{DD} = V_F + I_F R$$

V_{DD} and R are constants while V_F and I_F can change

OPERATING POINT...

- Recall equation of a straight line:

$y = ax + b$ where a and b are real values



You need only 2 point
to draw a straight line

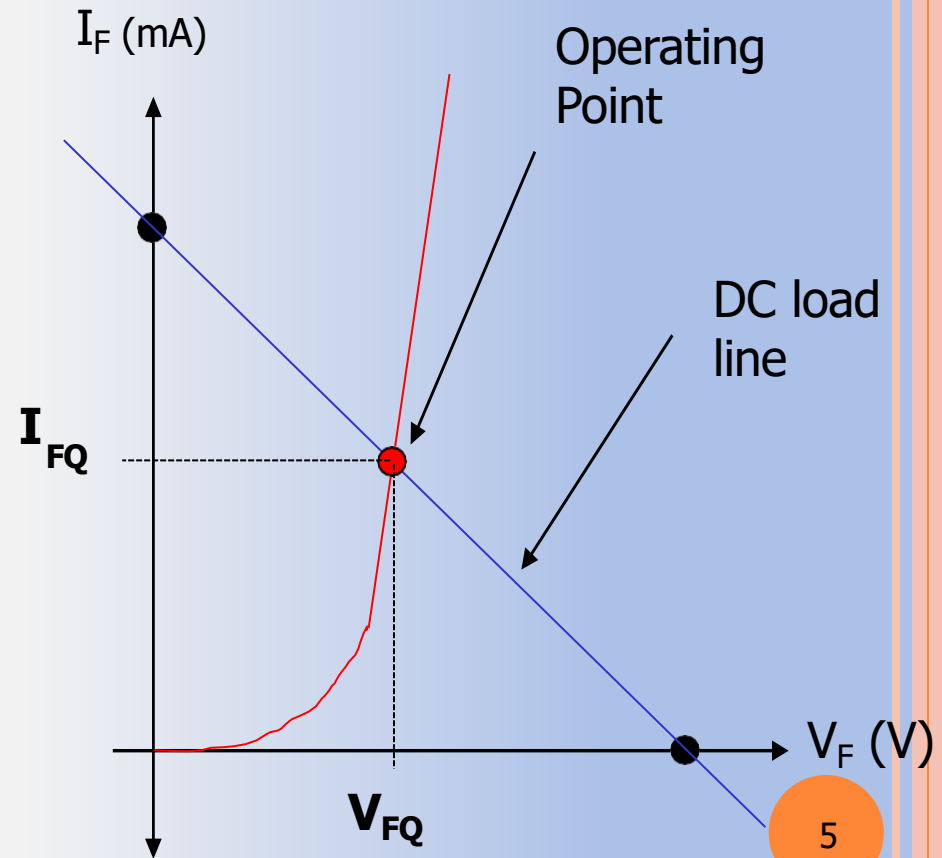
OPERATING POINT...

- Let's look at the dc load equation:

$$V_{DD} = V_F + I_F R$$

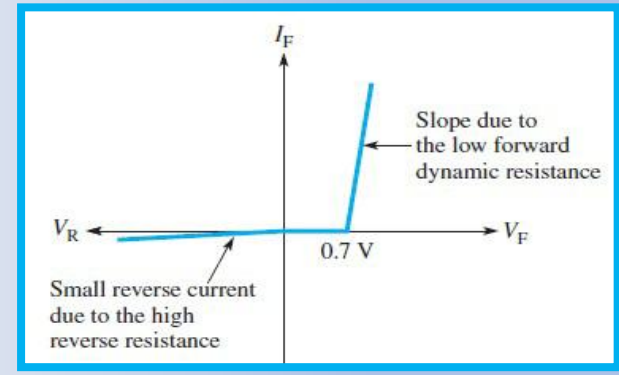
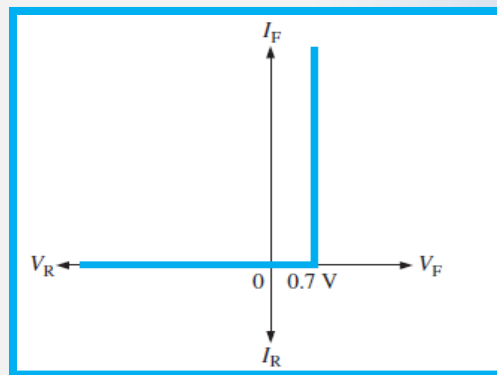
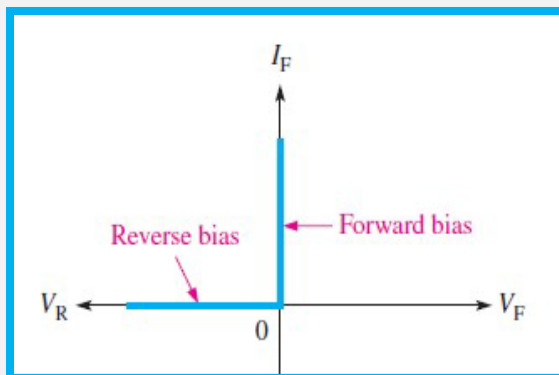
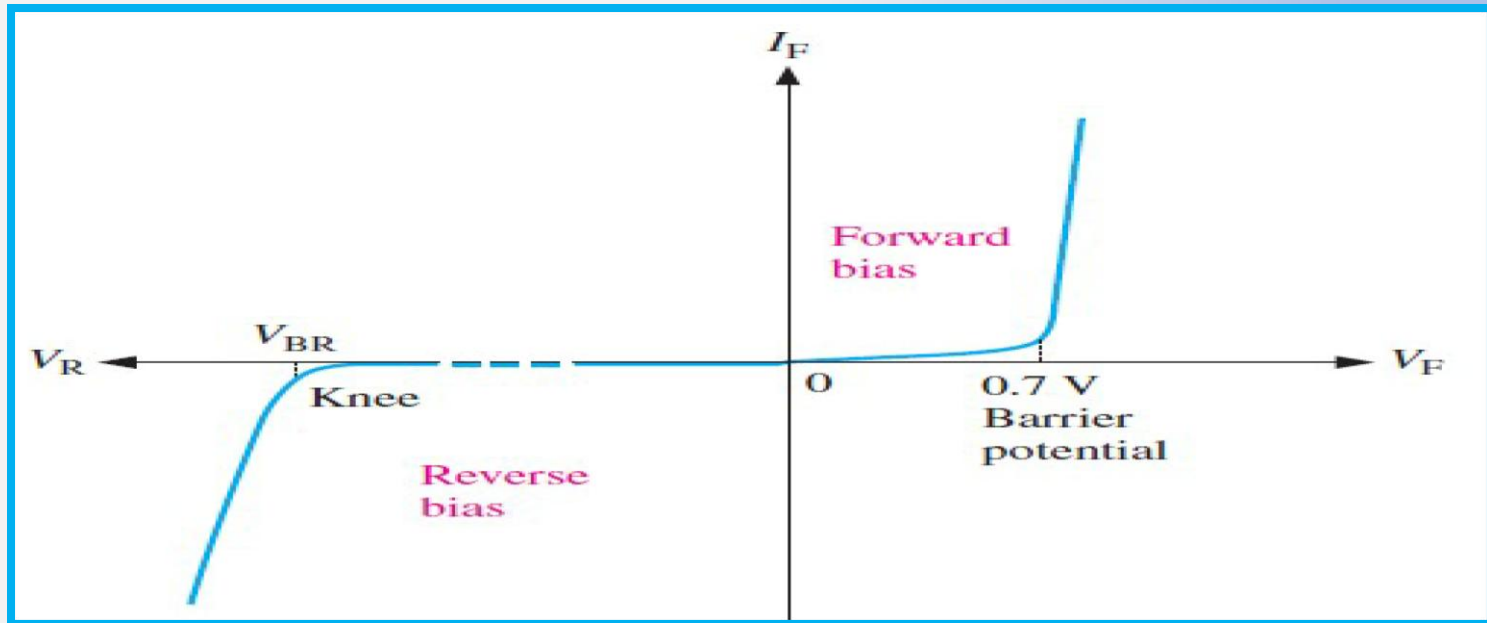
→ If $V_F = 0$, $I_F = V_{DD}/R$
 $(V_F, I_F) = (0, V_{DD}/R)$

→ If $I_F = 0$, $V_F = V_{DD}$
 $(V_F, I_F) = (V_{DD}, 0)$



Diode Models

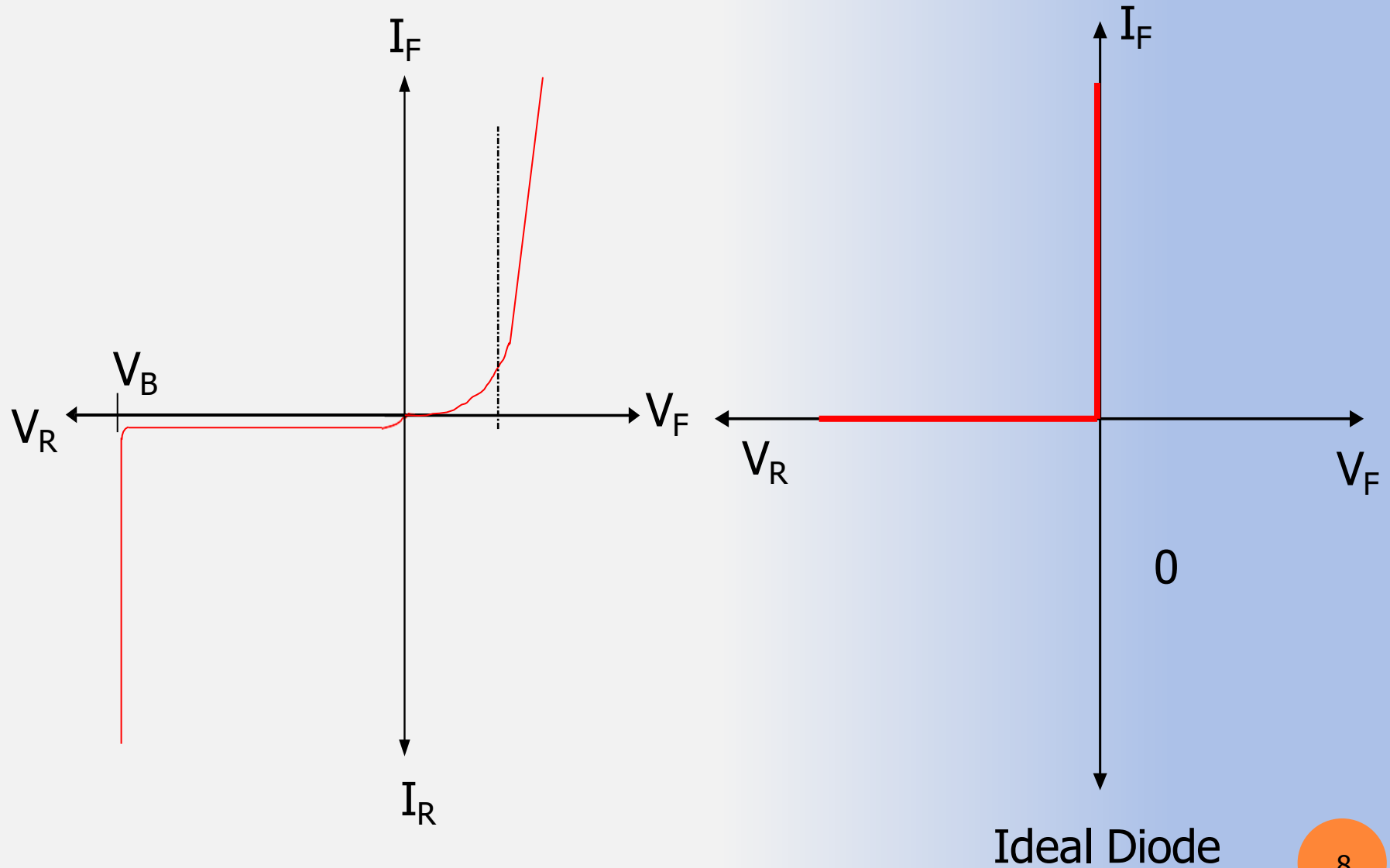
□ Diode $i-v$ characteristics equation



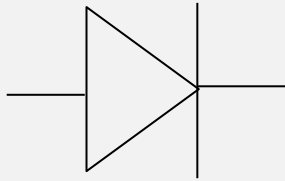
DIODE APPROXIMATIONS MODELS

- **Approximations** are used in electronics to **simplify analysis** of electronic circuits to avoid difficulty and time-consuming calculations
- First Approximation: **Ideal Diode**
 - ✓ Forward barrier voltage is ignored (0V)
 - ✓ Reverse saturation current is ignored (0A)
 - ✓ No breakdown voltage (infinity)
- In the forward bias the diode conducts at 0V

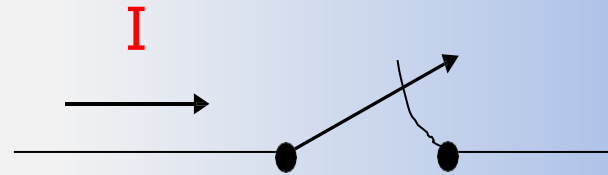
FIRST APPROXIMATIONS MODEL



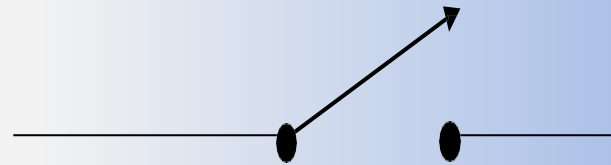
FIRST APPROXIMATION...



Forward Biased Diode



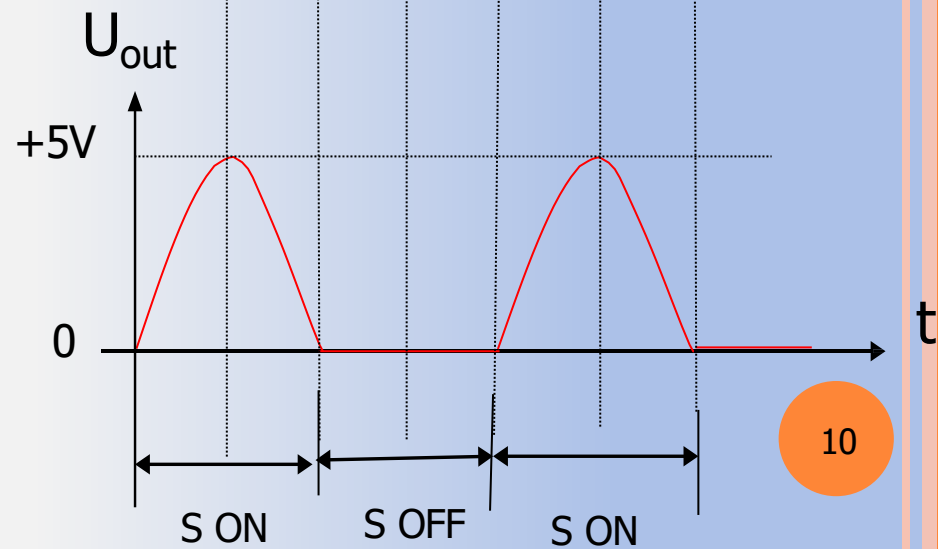
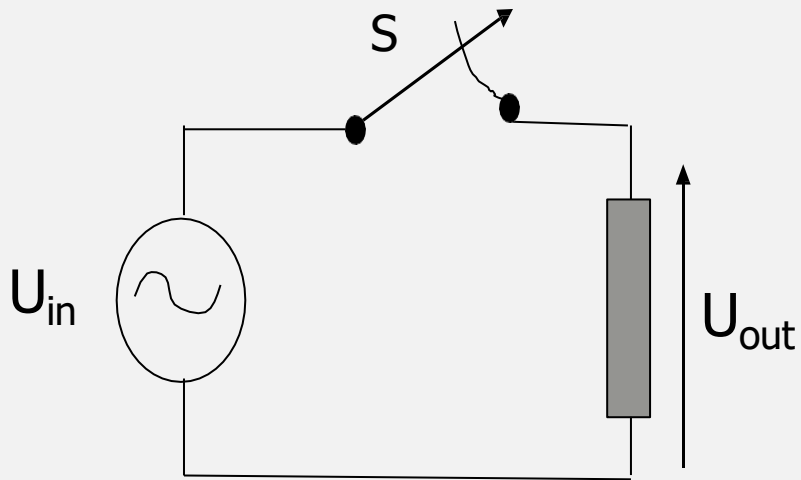
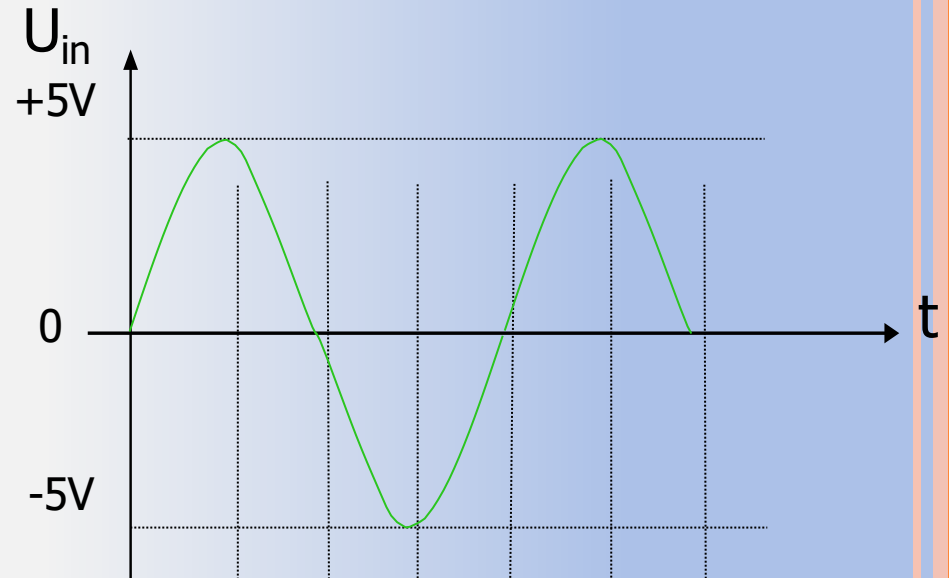
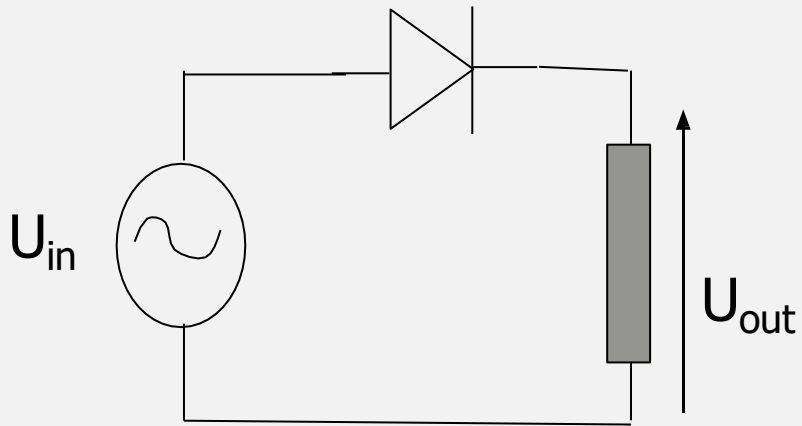
Equivalent: Closed Switch - Zero voltage drop



Reverse Biased Diode

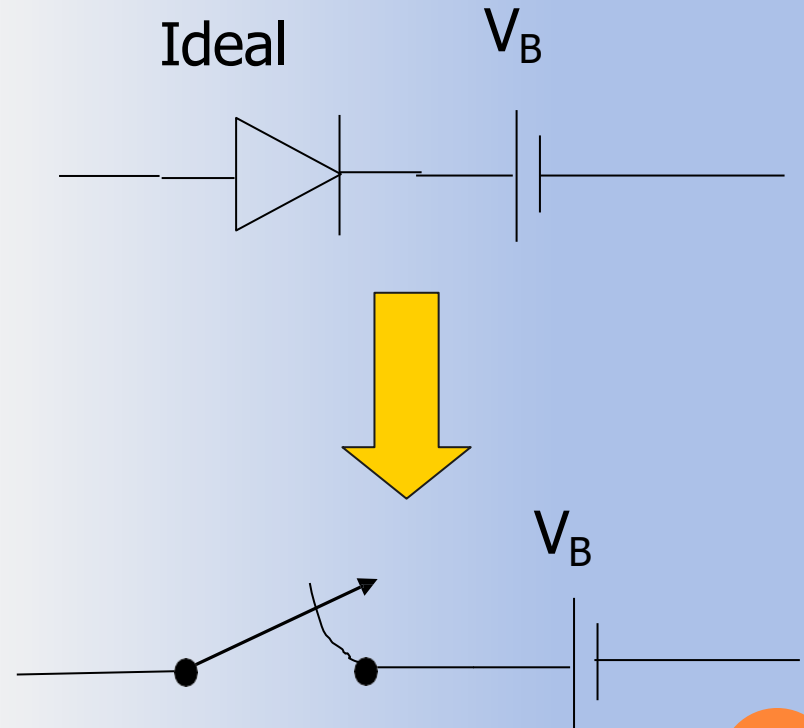
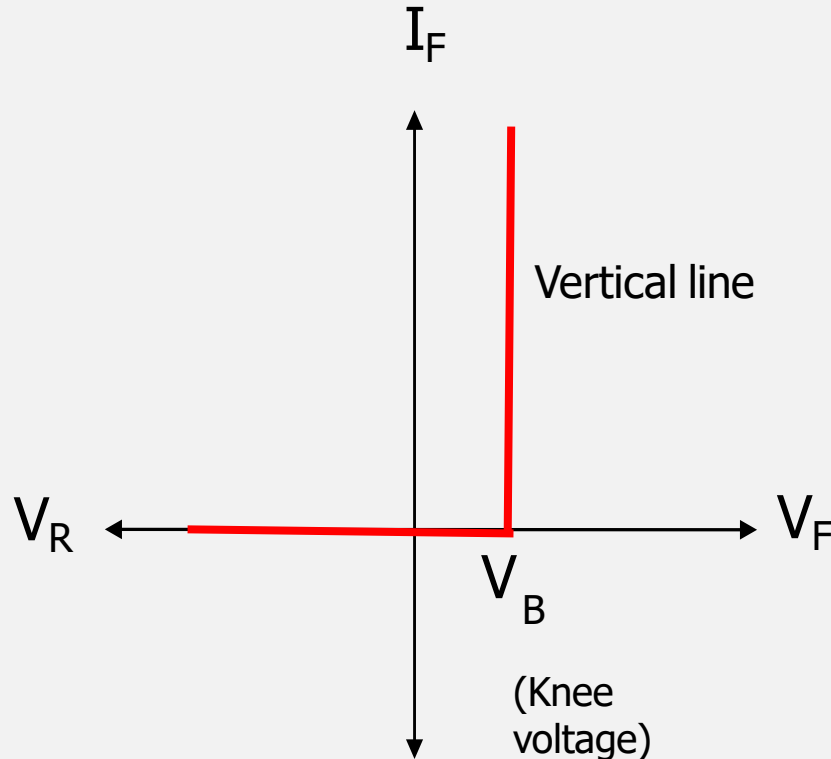
Equivalent: Open Switch - Infinite Resistance

FIRST APPROXIMATION...



SECOND APPROXIMATION MODEL

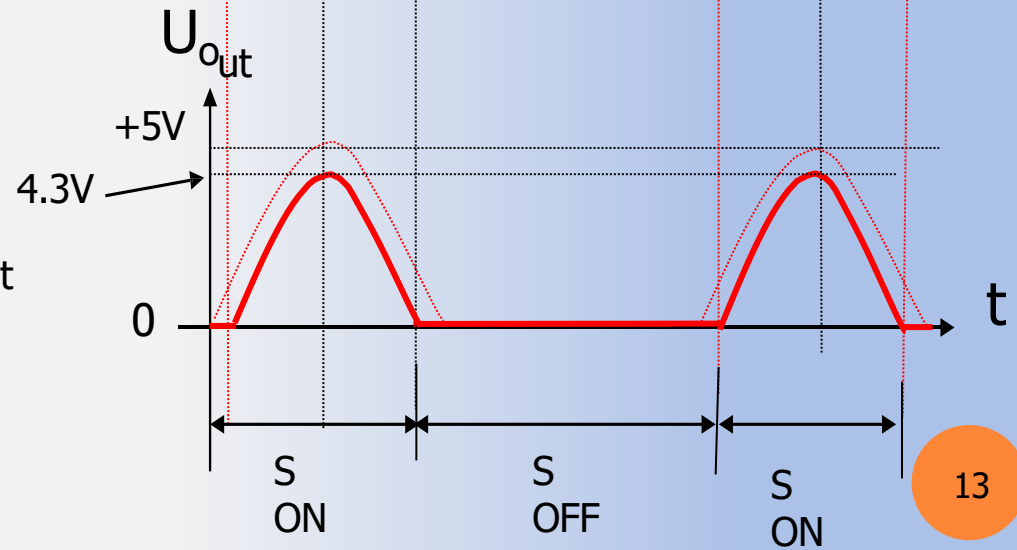
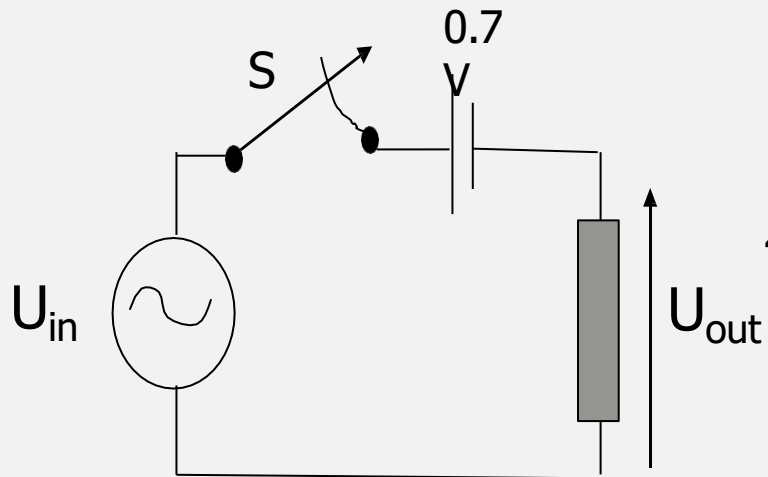
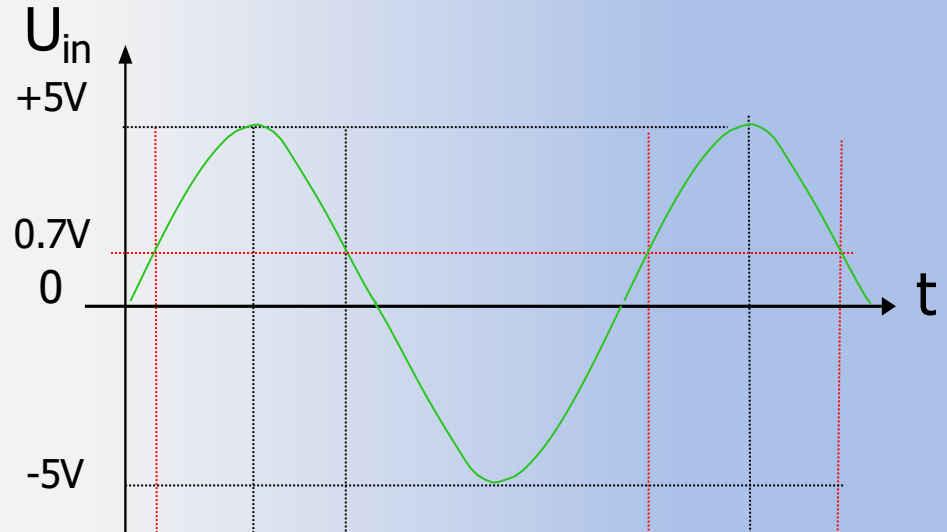
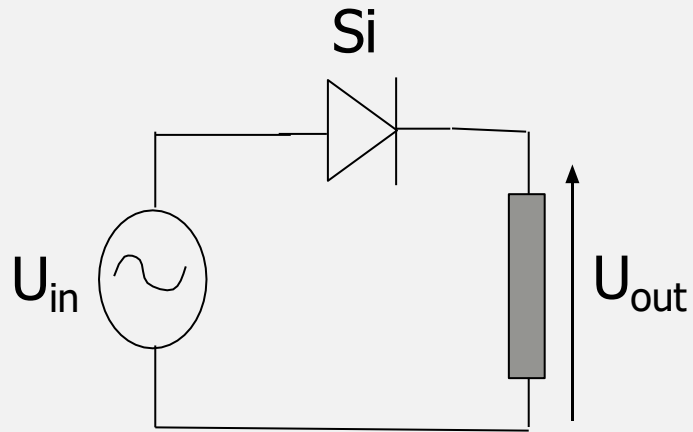
- The forward voltage drop across the real diode (Barrier voltage) is taken into consideration
- It is also called **Practical Diode Model**



SECOND APPROXIMATION...

- ❑ The diode will conduct in the forward direction if the external voltage exceeds the Barrier Voltage (V_B)
- ❑ The equivalent circuit is an ideal diode in series with a battery whose voltage equals the knee voltage (0.7V Si, 0.3V Ge)
- ❑ In the reverse direction, the diode is equivalent to an **open switch**

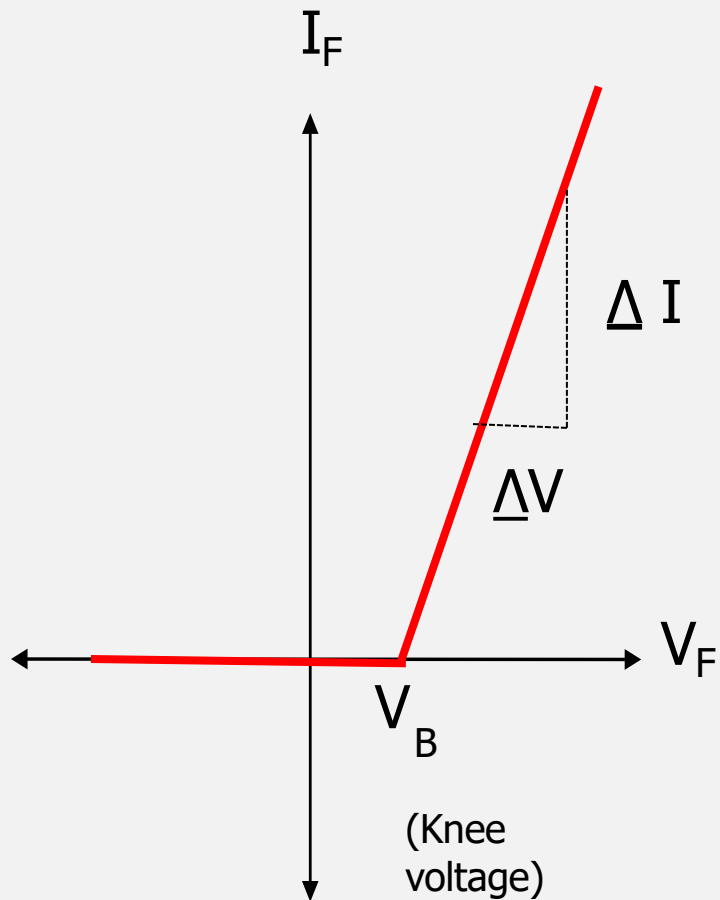
SECOND APPROXIMATION...



THIRD APPROXIMATION MODEL

- ❑ Third approximation can be used to account more accurately for diode forward voltage drop
- ❑ It is also called **the complete Diode Model**.
- ❑ The diode I-V characteristic is not vertical above the knee voltage but actually slopes upwards to the right
- ❑ This means that as more current flows through the real diode, more voltage is dropped across it
- ❑ Once the diode conducting, it acts like a resistor

THIRD APPROXIMATION....



Change in voltage is directly proportional to the change in current

$$r = \frac{\Delta V}{\Delta I}$$

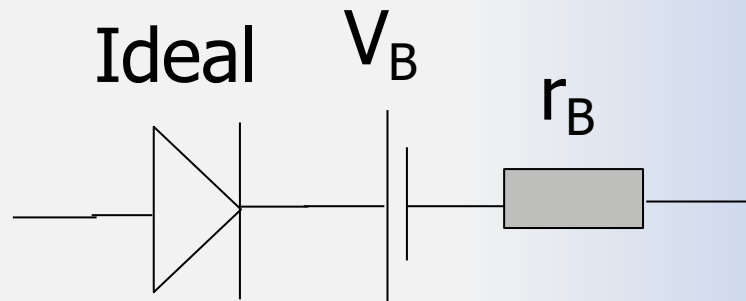
This is resistance of the semiconductor material (p and n regions)

This resistance is called **forward or bulk resistance of the diode (r_B)**

$$r_B = \frac{\Delta V}{\Delta I}$$

THIRD APPROXIMATION...

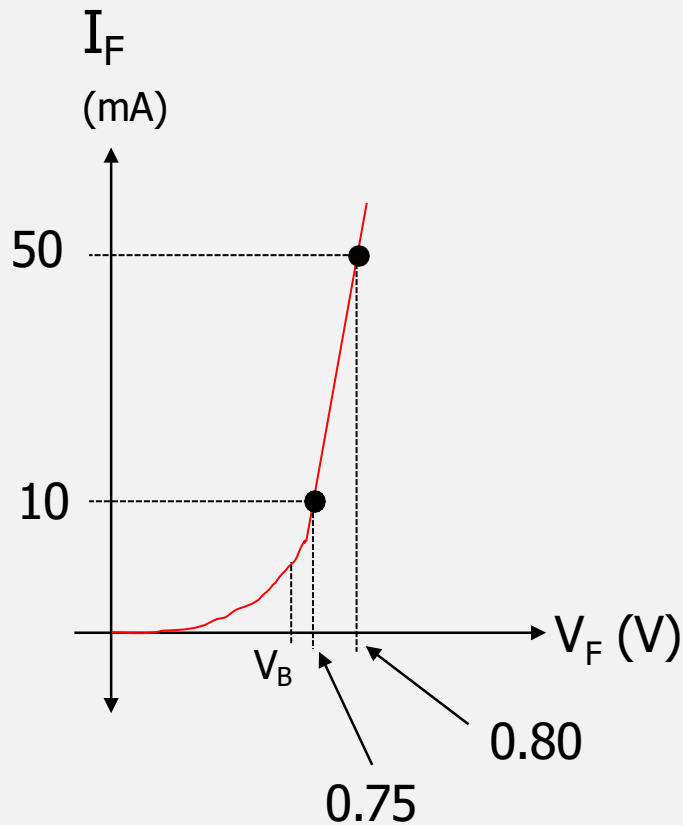
- The equivalent circuit is an ideal diode in series with a battery of value V_B and a resistor of value r_B



- It takes at least 0.3V or 0.7V to turn on the diode.
- The diode then acts like a resistor of value r_B which drops additional voltage depending on the amount of current drawn

ESTIMATING BULK RESISTANCE

If the I-V characteristic is available, select two points well above the knee of the curve



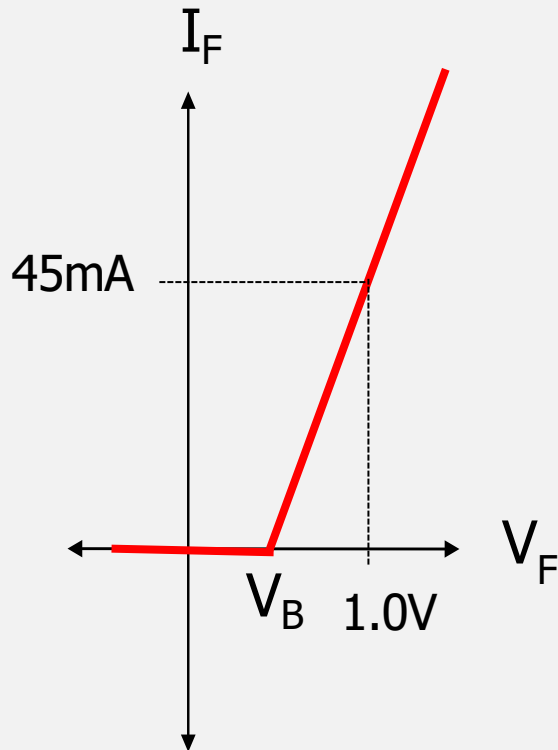
$$r_B = \frac{\Delta V}{\Delta I}$$

$$= \frac{0.8V - 0.75V}{50mA - 10mA} = \frac{0.05V}{40mA} = 1.2\Omega$$

ESTIMATING BULK RESISTANCE...

Read the forward current at 1V normally specified in Manufacturer's data sheet

Use the knee voltage and current to workout r_B



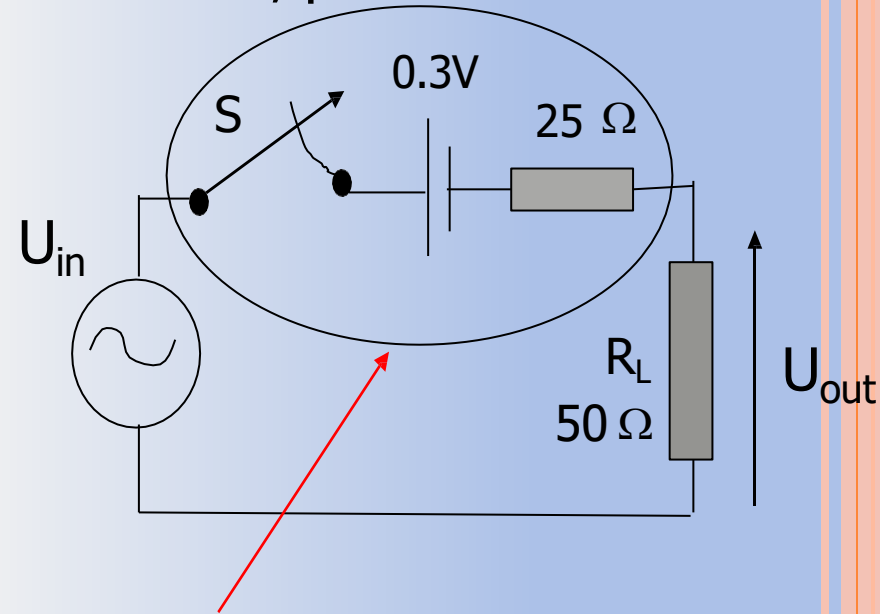
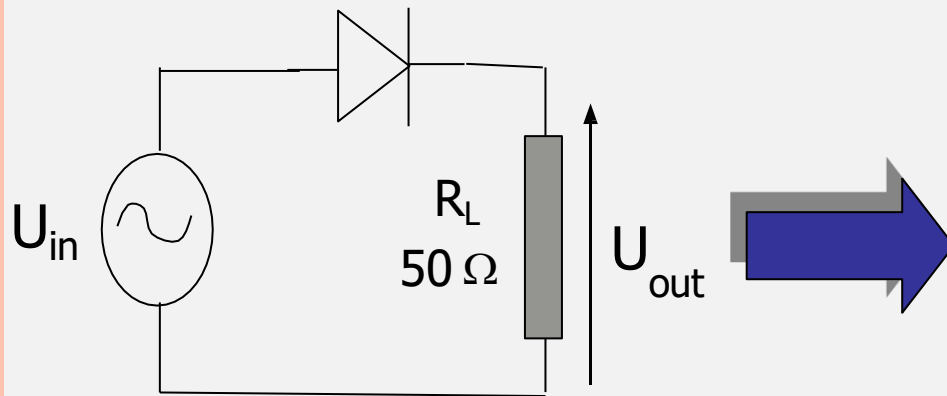
$$r_B = \frac{1V - V_B}{45mA - 0}$$

For Si, $V_B = 0.7V$, and for Ge, $V_B = 0.3V$

THIRD APPROXIMATION...

Example:

Use third approximation to sketch U_{out} waveform, if the diode is made of germanium and it has I_F of 28mA at 1V, peak $U_{in} = 2V$



Diode equivalent circuit
using 3rd Approximation

$$r_B = \frac{1V - 0.3V}{28mA} = 25\Omega$$

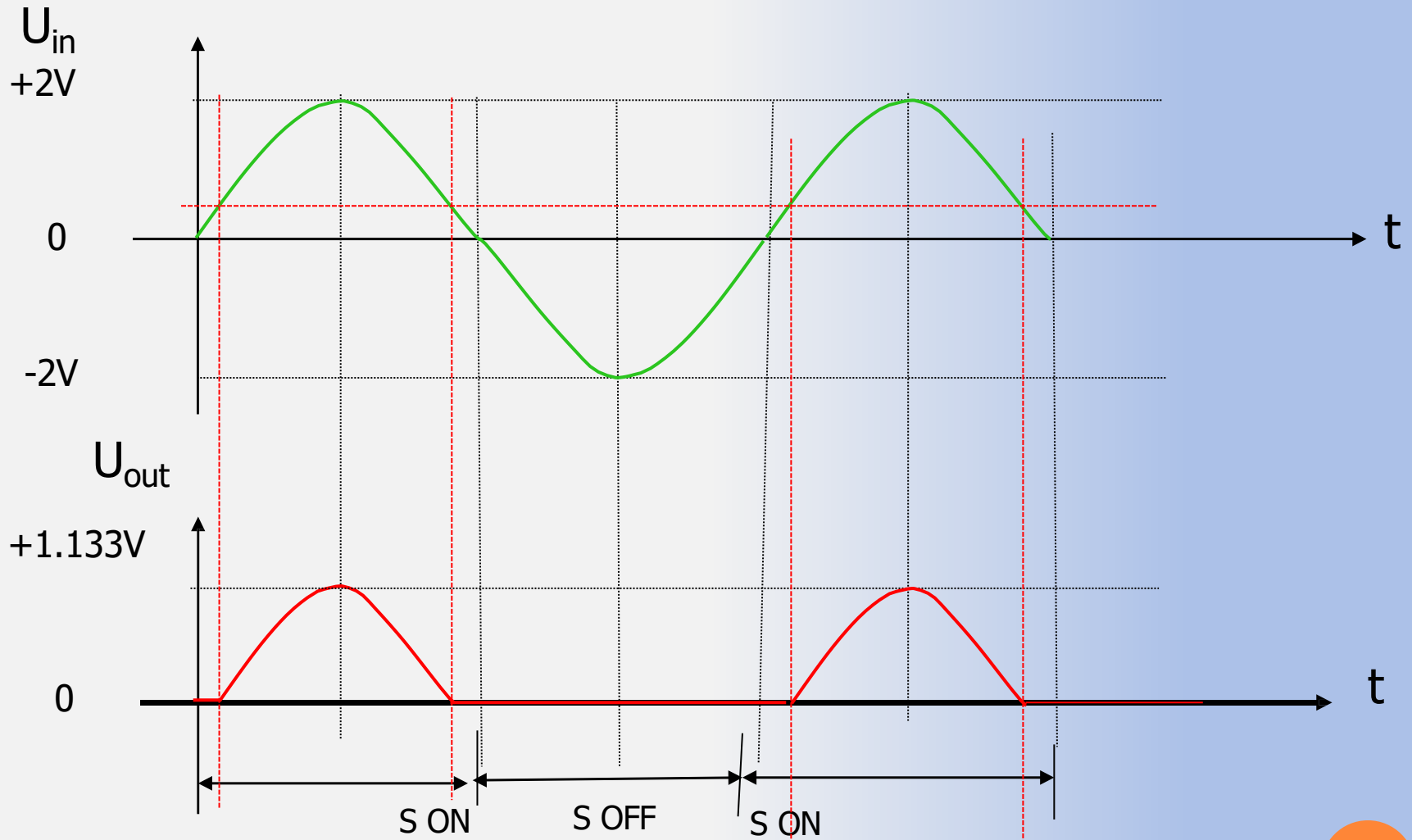
THIRD APPROXIMATION...

- Determine the peak U_{out}

$$\begin{aligned}U_{out_p} &= \frac{(U_{in_p} - V_B)R_L}{R_L + r_B} \\&= \frac{(2V - 0.3V)50\Omega}{50\Omega + 25\Omega} = 1.133V\end{aligned}$$

- The diode will conduct after the U_{in} has reached $0.3V$

THIRD APPROXIMATION...



USING THE DIODE APPROXIMATIONS

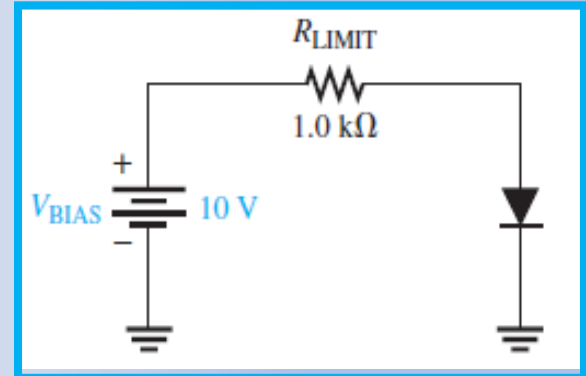
- ❑ Start with **the ideal-diode** approach to get basic idea of how the circuit operates
- ❑ If the answers obtained by the ideal-diode approach indicate that 0.3V or 0.7V is significant, **use the second approximation**
- ❑ If the external resistance in series with the diode is not large compared with the bulk resistance of the diode, **use the third approximation**

DIODE APPROXIMATIONS...

- ❑ The approximations we have studied are called **Large Signal Diode Approximation**
- ❑ It is called Large Signal approximation **because** the circuits are driven by large-signal sources whose **voltages are much higher** than the diode **knee voltage**
- ❑ Therefore, the approximations are suitable whenever the signal driving the diode is larger than the knee voltage

Diode Models

□ **Example:** Determine the forward voltage and forward current for the diode in Figure for each of the diode models. Also find the voltage across the limiting resistor in each case. Assume at the determined value of forward current.



Solution Ideal model:

$$V_F = 0 \text{ V}$$

$$I_F = \frac{V_{BIAS}}{R_{LIMIT}} = \frac{10 \text{ V}}{1.0 \text{ k}\Omega} = 10 \text{ mA}$$

$$V_{R_{LIMIT}} = I_F R_{LIMIT} = (10 \text{ mA})(1.0 \text{ k}\Omega) = 10 \text{ V}$$

Practical model:

$$V_F = 0.7 \text{ V}$$

$$I_F = \frac{V_{BIAS} - V_F}{R_{LIMIT}} = \frac{10 \text{ V} - 0.7 \text{ V}}{1.0 \text{ k}\Omega} = \frac{9.3 \text{ V}}{1.0 \text{ k}\Omega} = 9.3 \text{ mA}$$

$$V_{R_{LIMIT}} = I_F R_{LIMIT} = (9.3 \text{ mA})(1.0 \text{ k}\Omega) = 9.3 \text{ V}$$

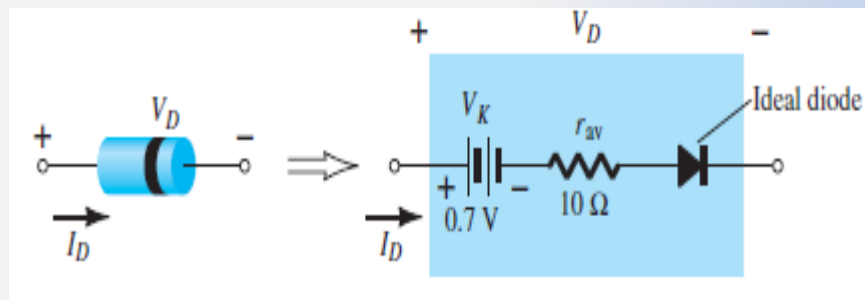
Complete model:

$$I_F = \frac{V_{BIAS} - 0.7 \text{ V}}{R_{LIMIT} + r'_d} = \frac{10 \text{ V} - 0.7 \text{ V}}{1.0 \text{ k}\Omega + 10 \Omega} = \frac{9.3 \text{ V}}{1010 \Omega} = 9.21 \text{ mA}$$

$$V_F = 0.7 \text{ V} + I_F r'_d = 0.7 \text{ V} + (9.21 \text{ mA})(10 \Omega) = 792 \text{ mV}$$

$$V_{R_{LIMIT}} = I_F R_{LIMIT} = (9.21 \text{ mA})(1.0 \text{ k}\Omega) = 9.21 \text{ V}$$

Diode Models

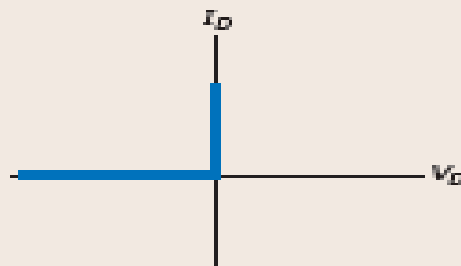


Summary

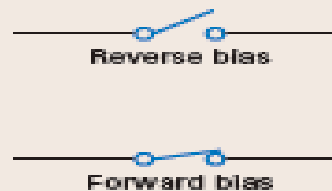
Diode Approximations

First or ideal

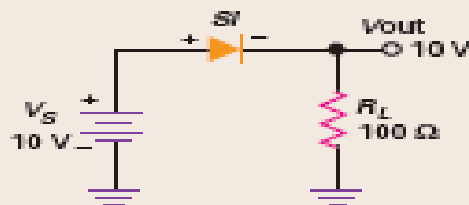
Diode curve



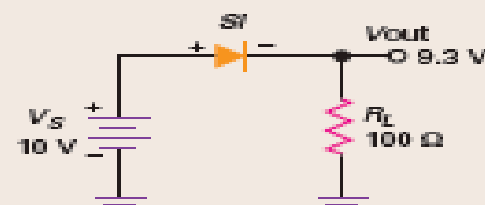
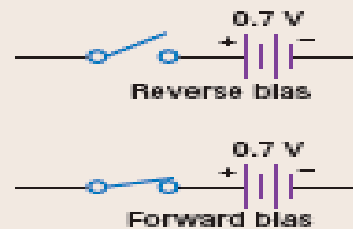
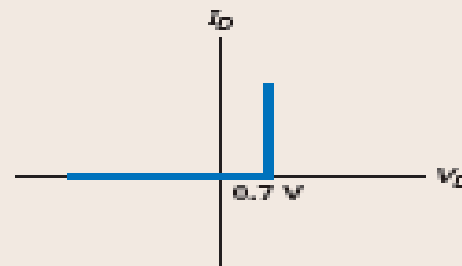
Equivalent circuit



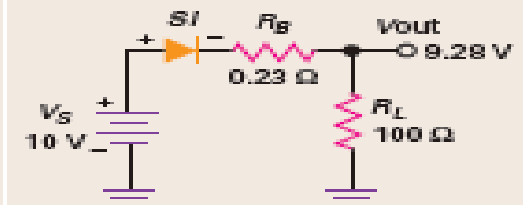
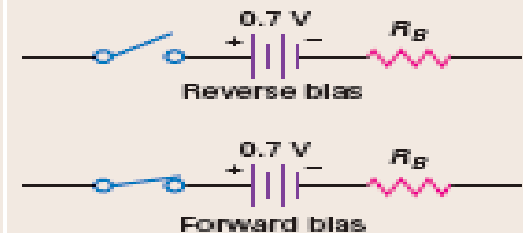
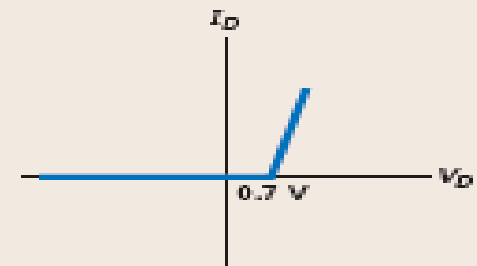
Circuit example



Second or practical

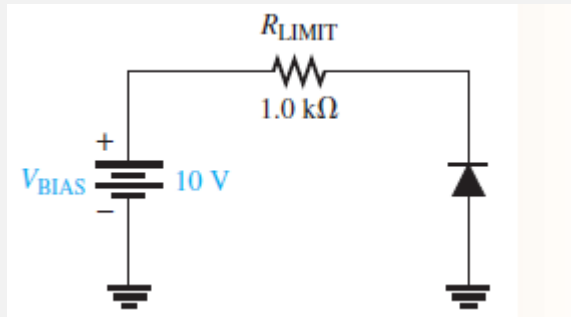


Third



Diode Models

- **Example:** Determine the reverse voltage and reverse current for the diode in Figure for each of the diode models. Also find the voltage across the limiting resistor in each case. Assume



Ideal model:

$$\begin{aligned} I_R &= 0 \text{ A} \\ V_R &= V_{\text{BIAS}} = 10 \text{ V} \\ V_{R_{\text{LIMIT}}} &= 0 \text{ V} \end{aligned}$$

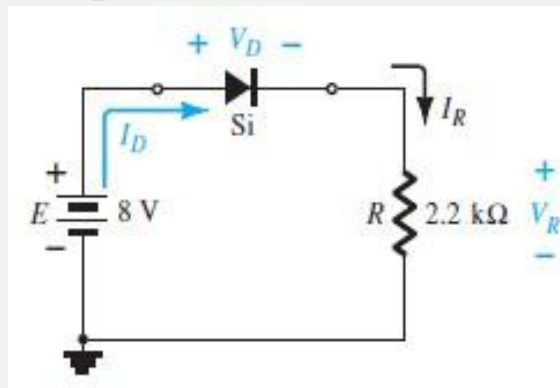
Practical model:

$$\begin{aligned} I_R &= 0 \text{ A} \\ V_R &= V_{\text{BIAS}} = 10 \text{ V} \\ V_{R_{\text{LIMIT}}} &= 0 \text{ V} \end{aligned}$$

Complete model:

$$\begin{aligned} I_R &= 1 \mu\text{A} \\ V_{R_{\text{LIMIT}}} &= I_R R_{\text{LIMIT}} = (1 \mu\text{A})(1.0 \text{ k}\Omega) = 1 \text{ mV} \\ V_R &= V_{\text{BIAS}} - V_{R_{\text{LIMIT}}} = 10 \text{ V} - 1 \text{ mV} = 9.999 \text{ V} \end{aligned}$$

- **Example:** For the series diode configuration of Fig. below, determine V_D , V_R , and I_D .

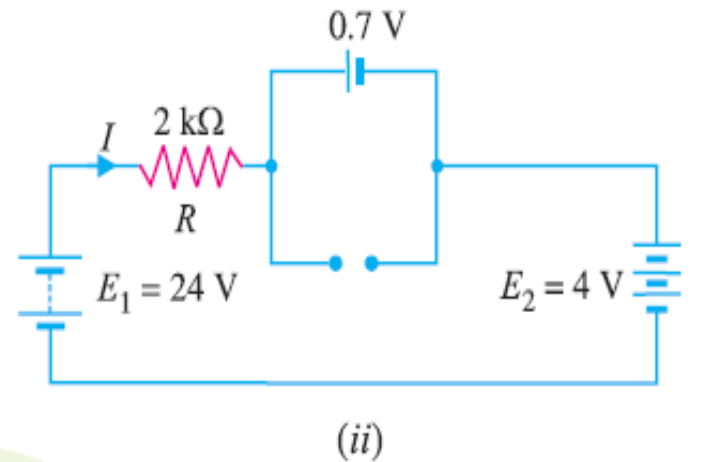
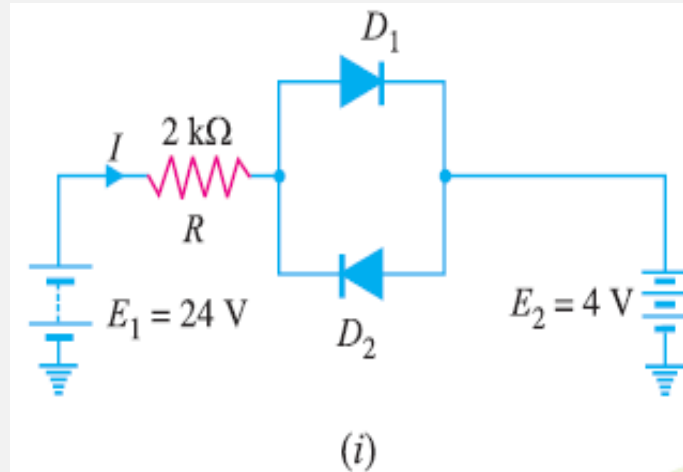


Solution:

$$\begin{aligned} V_D &= 0.7 \text{ V} \\ V_R &= E - V_D = 8 \text{ V} - 0.7 \text{ V} = 7.3 \text{ V} \\ I_D = I_R &= \frac{V_R}{R} = \frac{7.3 \text{ V}}{2.2 \text{ k}\Omega} \cong 3.32 \text{ mA} \end{aligned}$$

Diode Models

- Determine the current I in the circuit shown in Fig. below (i). Assume the diodes to be of silicon and forward resistance of diodes to be zero.



□ **Solution :**

The conditions of the problem suggest that diode D_1 is forward biased and diode D_2 is reverse biased. We can, therefore, consider the branch containing diode D_2 as open as shown in Fig. (ii). Further, diode D_1 can be replaced by its simplified equivalent circuit.

$$I = \frac{E_1 - E_2 - V_0}{R} = \frac{24 - 4 - 0.7}{2 \text{ k}\Omega} = \frac{19.3 \text{ V}}{2 \text{ k}\Omega} = 9.65 \text{ mA}$$

Assignment

Diode Models

- Find the voltage V_A in the circuit shown in Fig. 1. Use simplified model.
- Determine current through each diode in the circuit shown in Fig. 2. Use simplified model. Assume diodes to be similar.

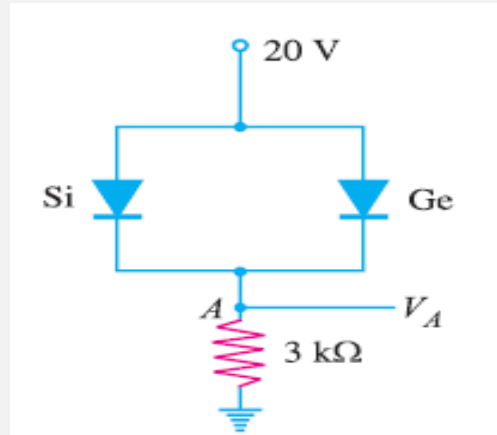


Fig. 1

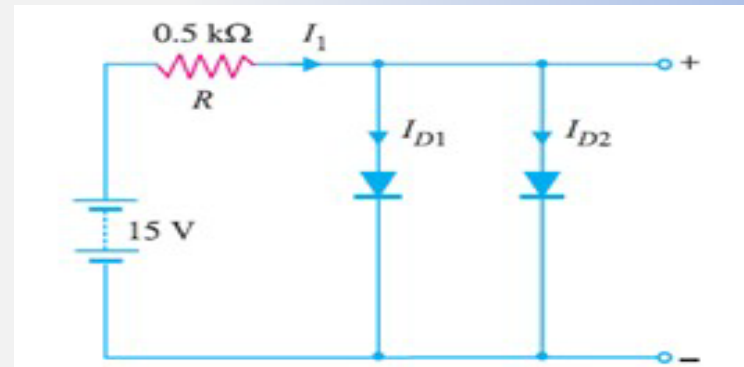
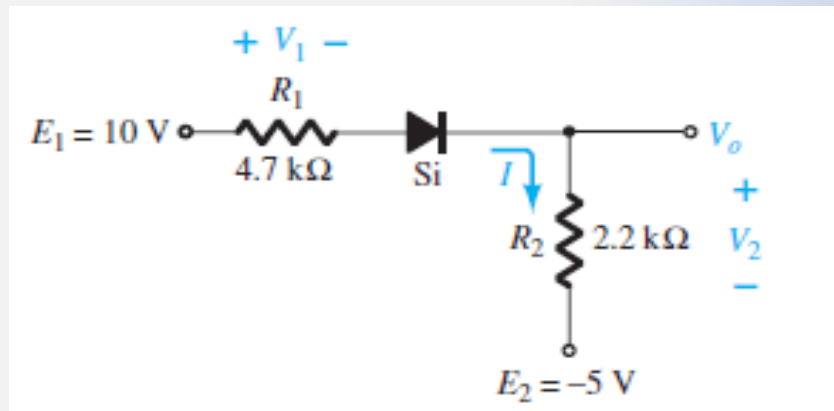


Fig. 2

- Determine I , V_1 , V_2 , and V_o for the series dc configuration of Fig. below .



Semiconductor Diodes and Applications

Diode Applications

Rectifiers

Clippers

Clampers

Detector

Regulators

Voltage -multiplier

Switches



Uses

- 0 Block current flow in a specific director.
- 0 Rectifier (AC to DC conversion).
- 0 Voltage regulator.
- 0 Limiter (protection).
- 0 Light Emitting Diodes (LEDs).
- 0 AM Detectors.
- 0 Electronic tuners.
- 0 Photodiodes



Diode



LED



Photo
Diode



Zener
Diode



Variable Capacity
(Varicap) Diode

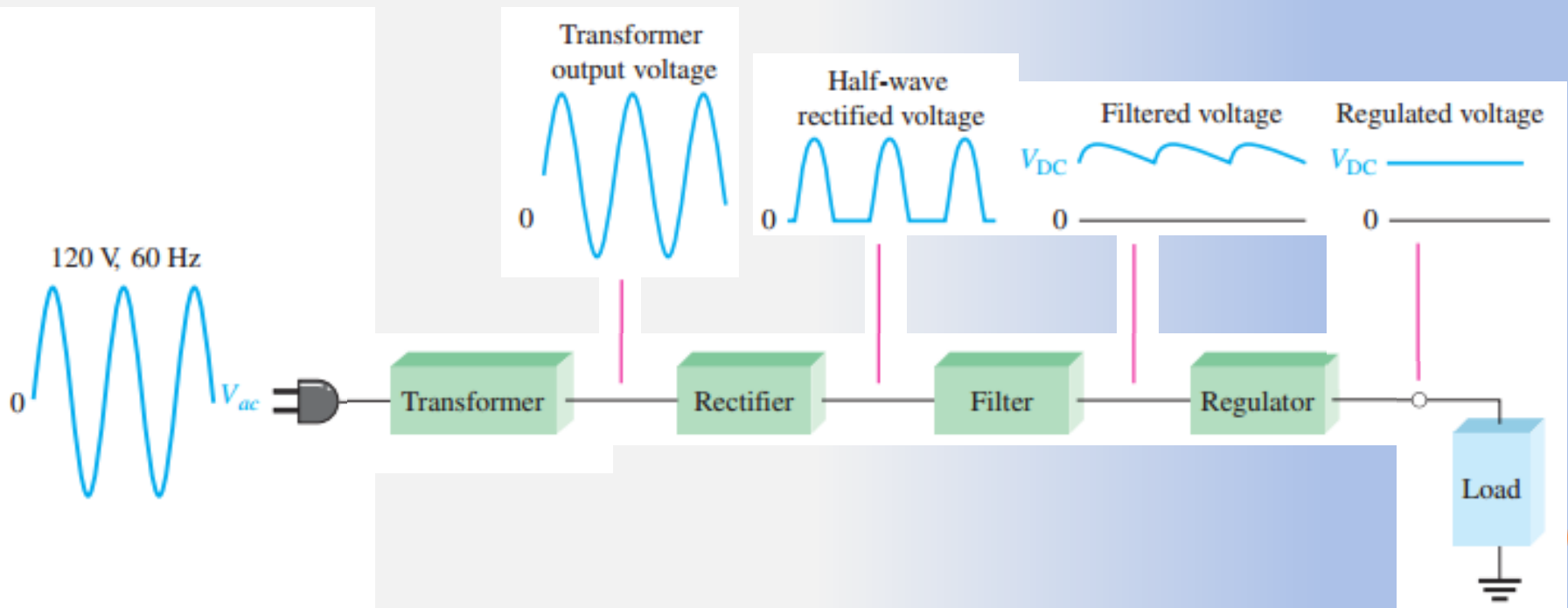


Schottky
Diode

Applications

1. Rectifier

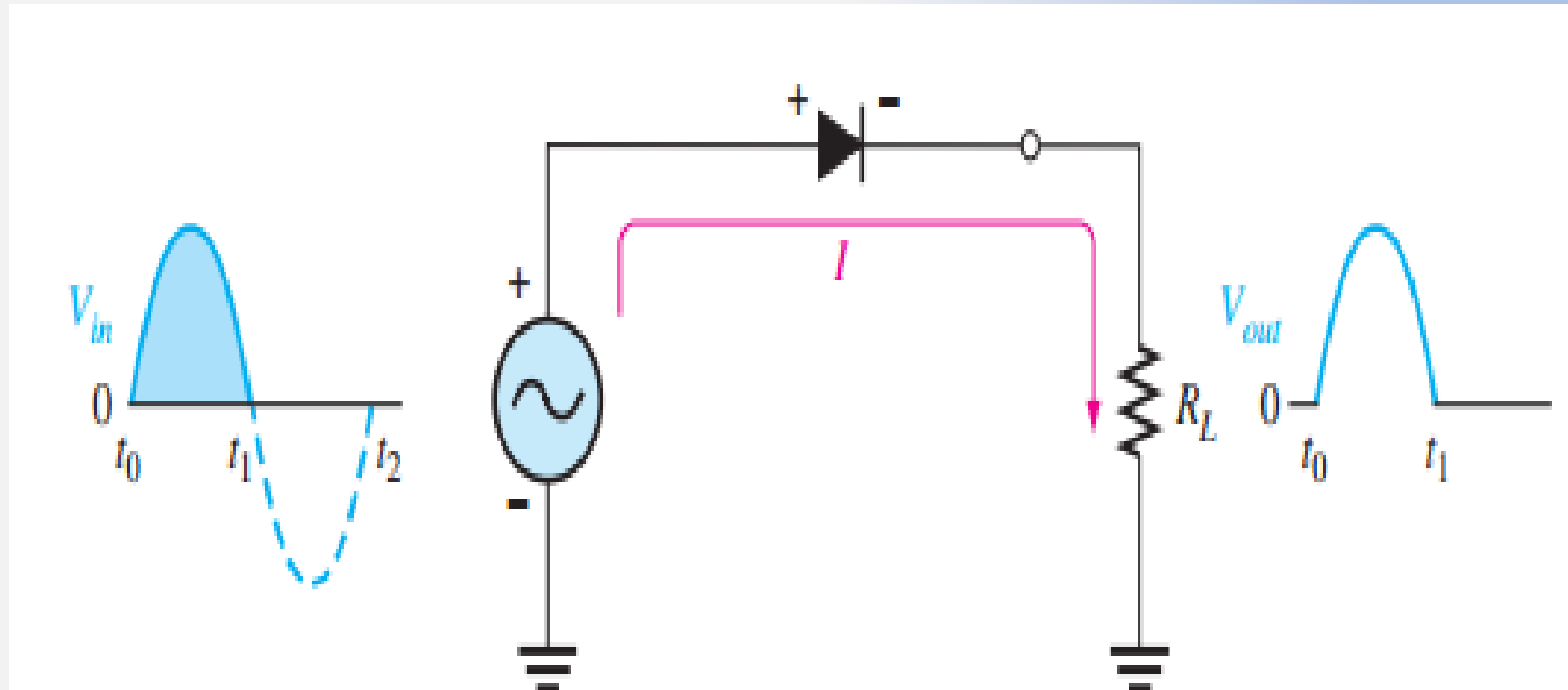
- Converts AC to DC
- Typically seen in DC Power supply



Applications

HALF-WAVE RECTIFIERS

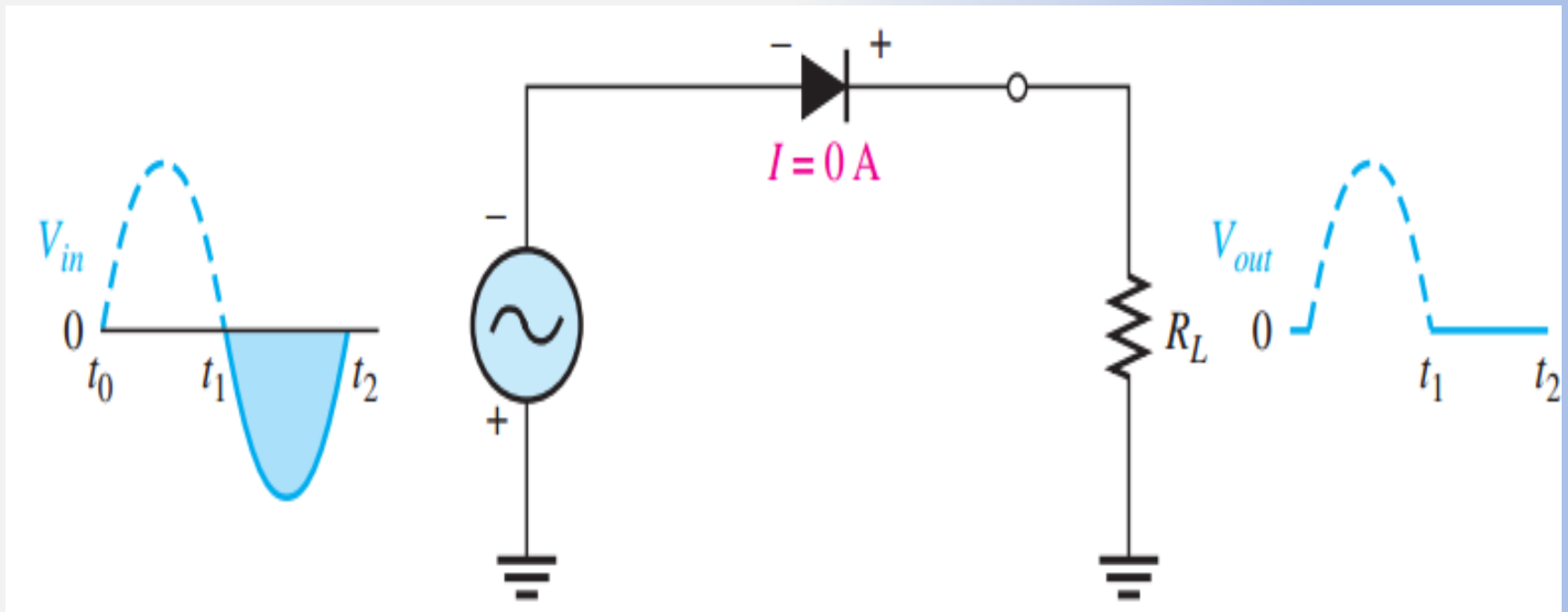
- During the positive alternation of the 50 Hz /60 Hz input voltage



Applications

HALF-WAVE RECTIFIERS

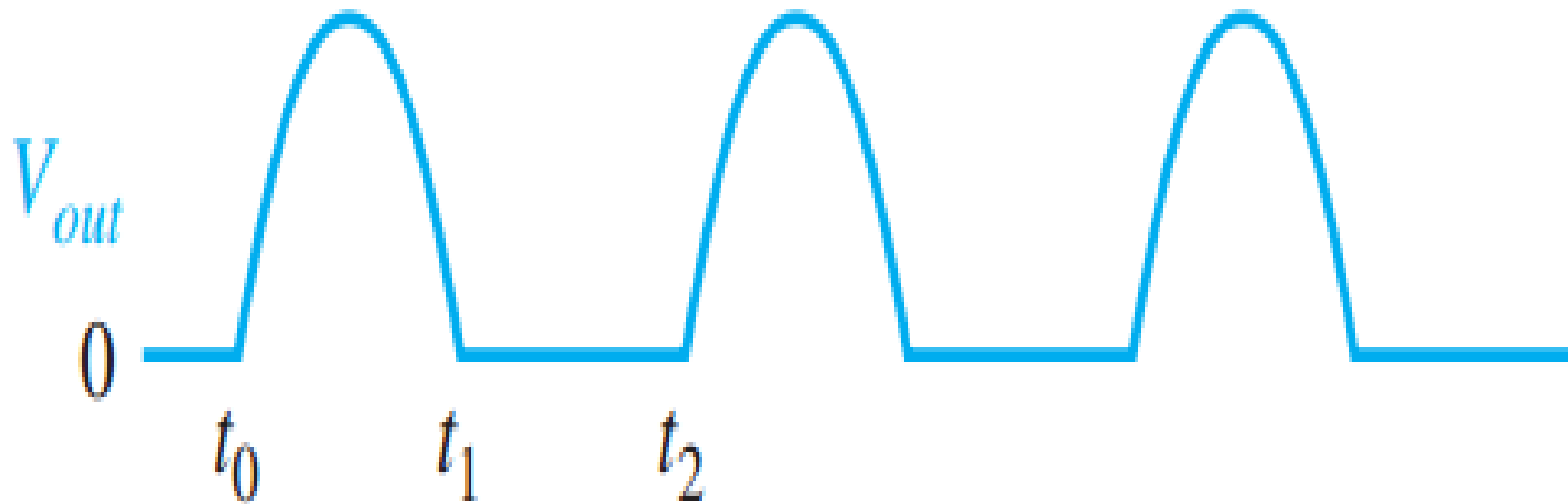
- During the negative alternation of the 50 Hz /60 Hz input voltage



Applications

HALF-WAVE RECTIFIERS

- Half-wave output voltage for three input cycles

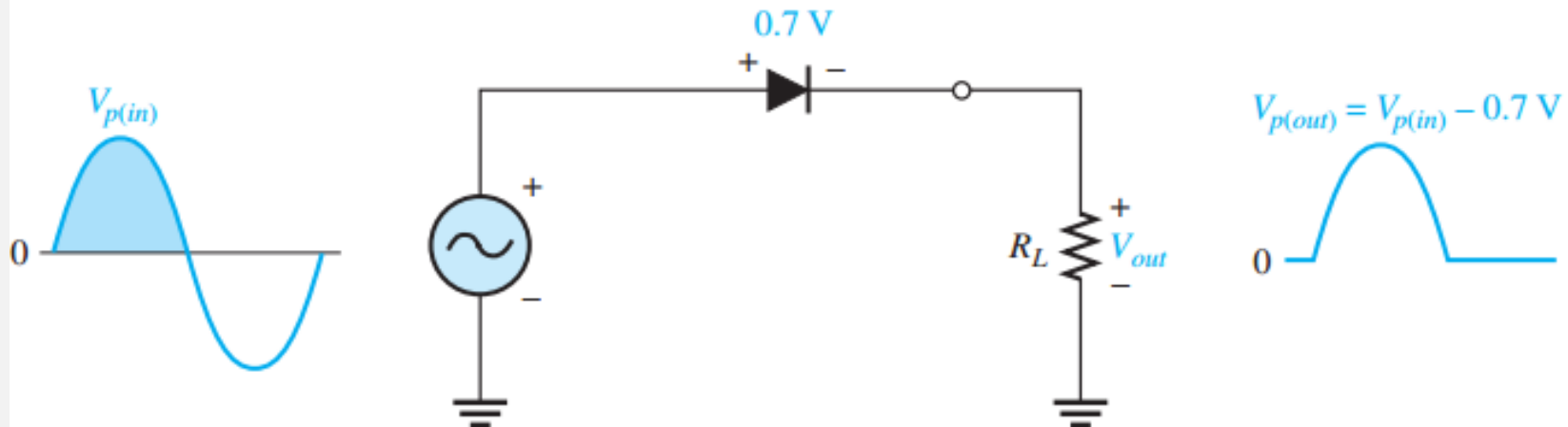


Applications

HALF-WAVE RECTIFIERS

- Effect of the Barrier Potential on the Half-Wave Rectifier Output

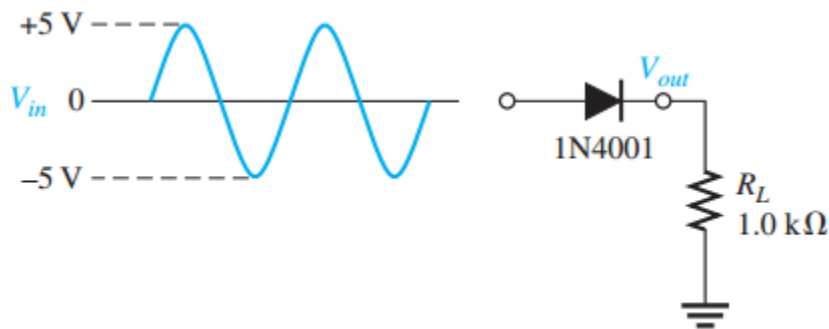
$$V_{p(out)} = V_{p(in)} - 0.7 \text{ V}$$



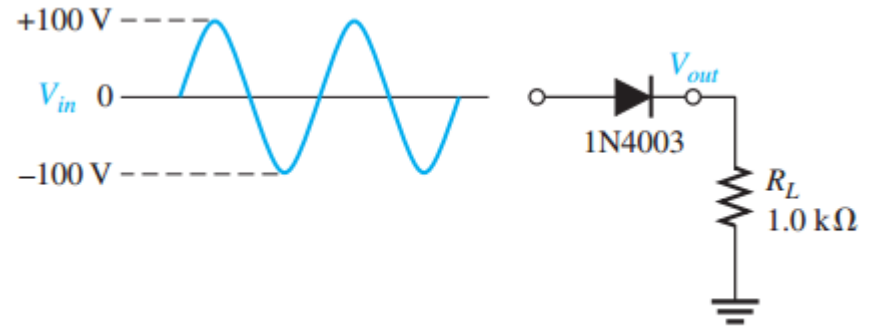
Applications

HALF-WAVE RECTIFIERS

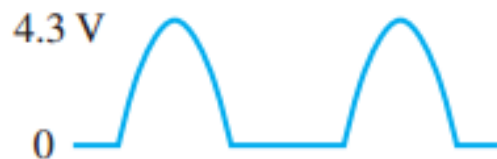
- Effect of the Barrier Potential on the Half-Wave Rectifier Output
- Example:** Draw the output voltages of each rectifier for the indicated input voltages



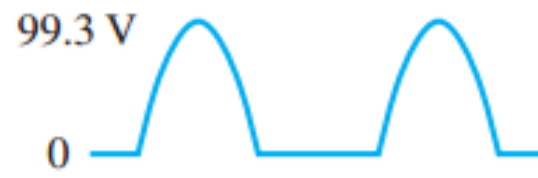
(a)



(b)



(a)



(b)

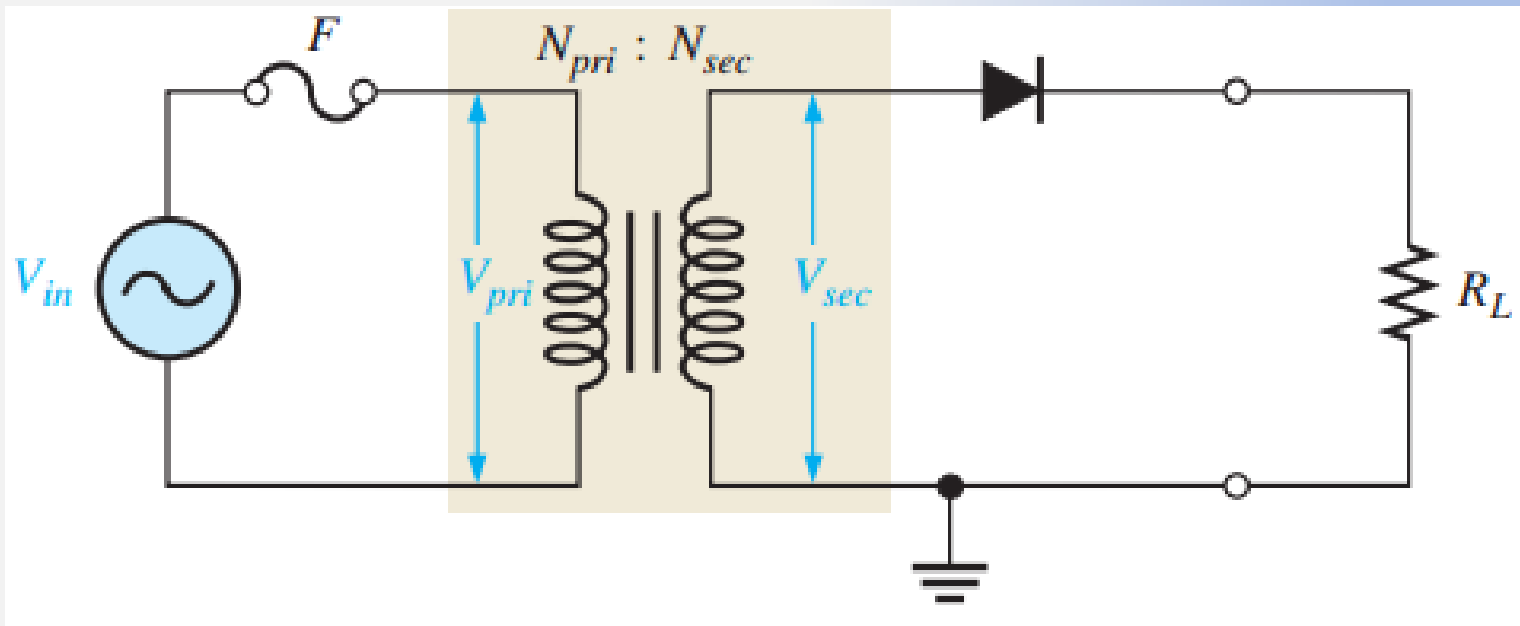
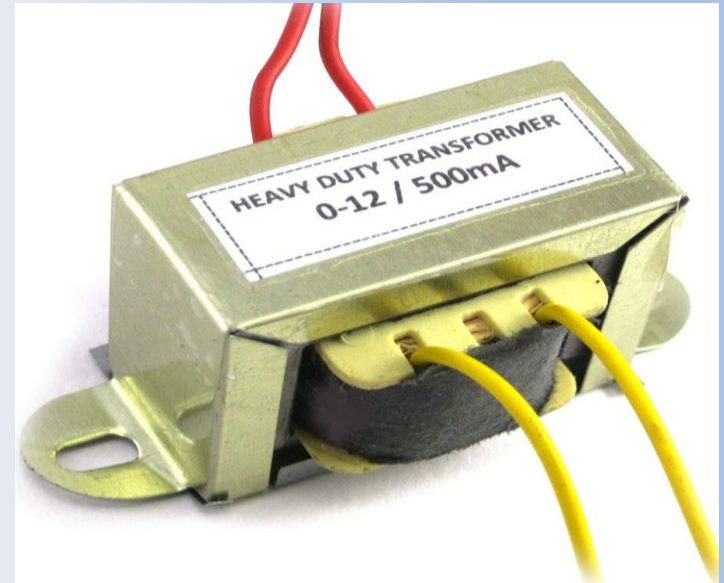


Applications

HALF-WAVE RECTIFIERS

○ Transformer Coupling:

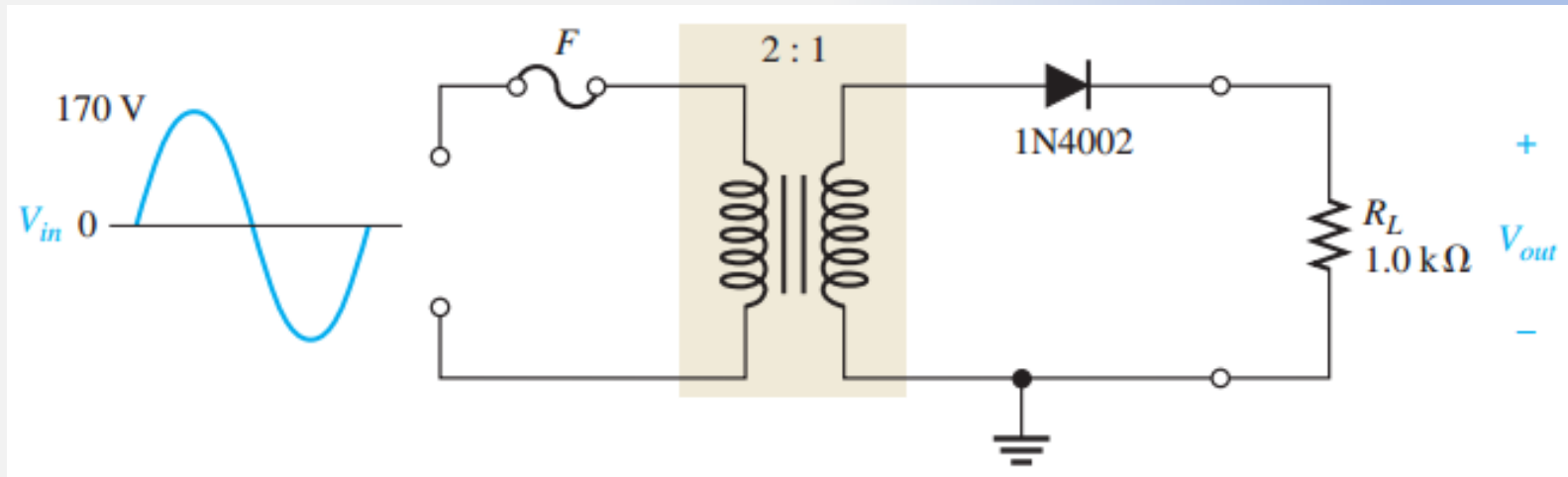
- stepdown
- isolation



Applications

HALF-WAVE RECTIFIERS

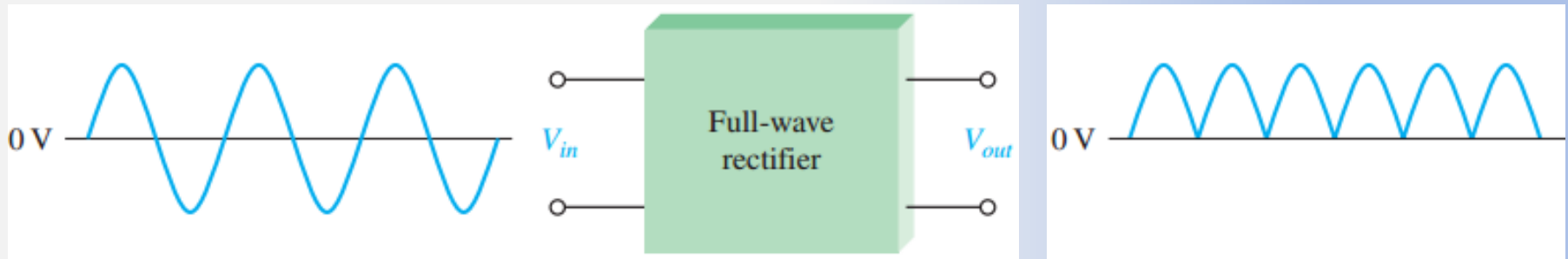
- Transformer Coupling:
- **Example:** Determine the peak value of the output voltage



Applications

FULL-WAVE RECTIFIERS

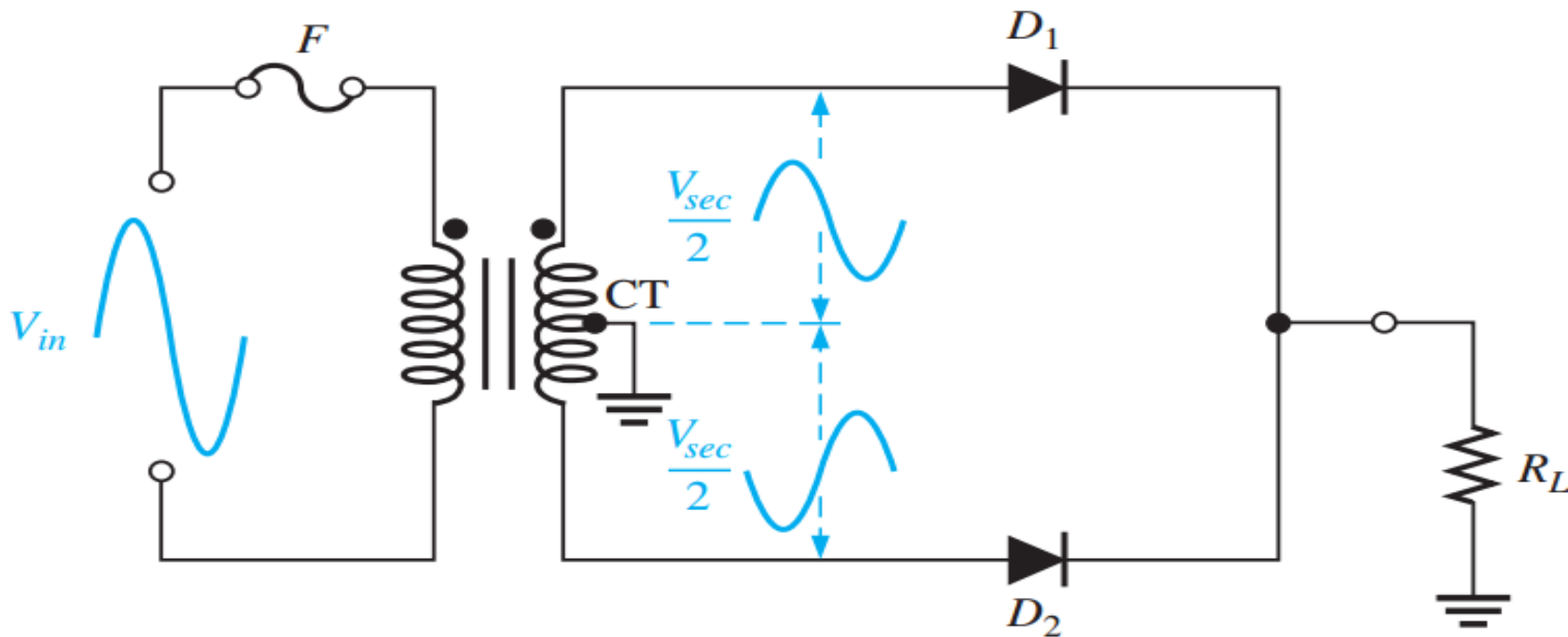
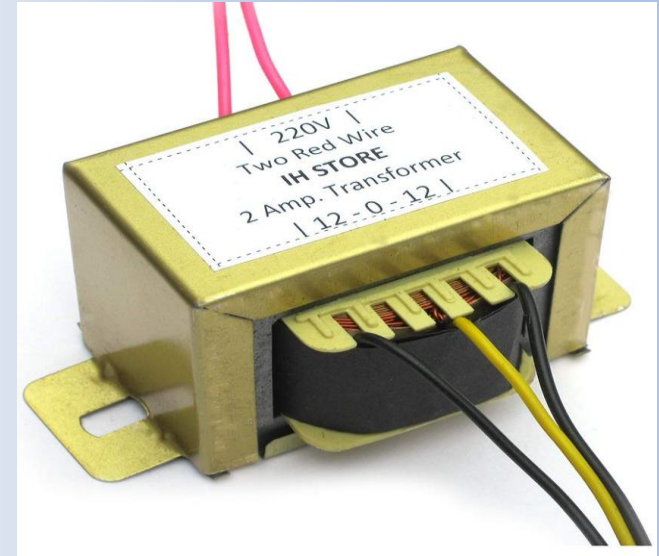
- Allows unidirectional (one-way) current through the load during the entire of the input cycle,
- Whereas a half-wave rectifier allows current through the load only during one-half of the cycle.
- The result of full-wave rectification is an output voltage with a frequency twice the input frequency and that pulsates every half-cycle of the input



Applications

FULL-WAVE RECTIFIERS

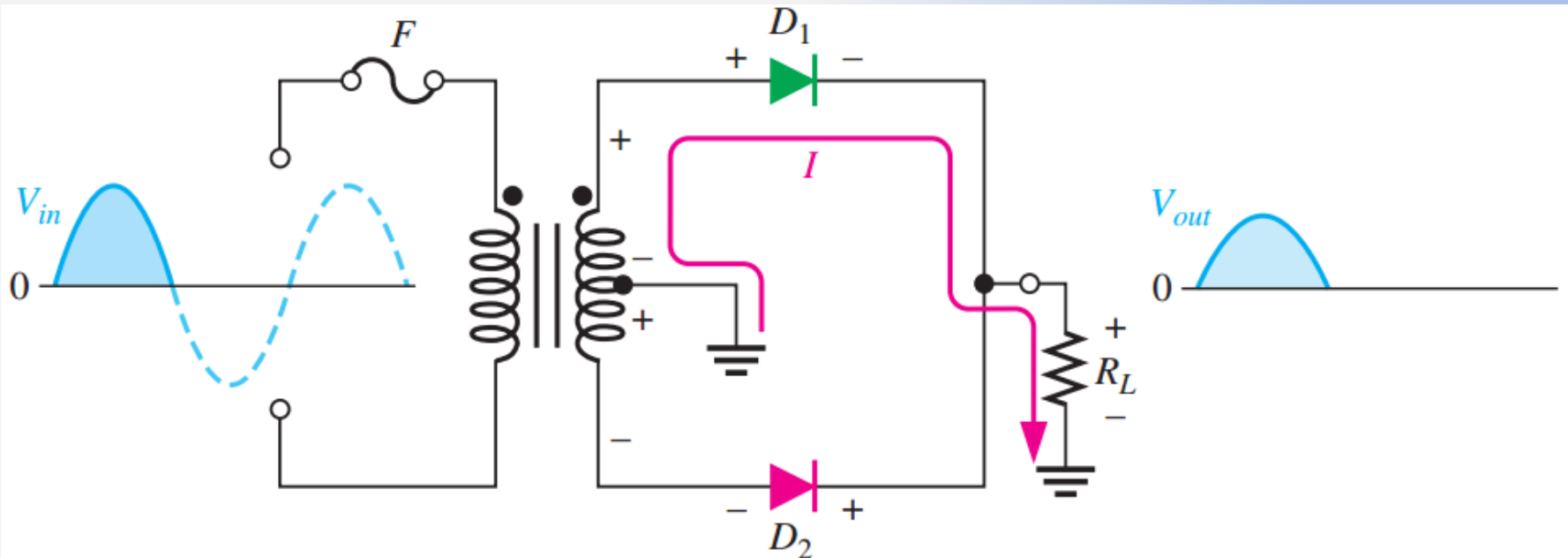
○ Center-Tapped Full-Wave Rectifier



Applications

FULL-WAVE RECTIFIERS

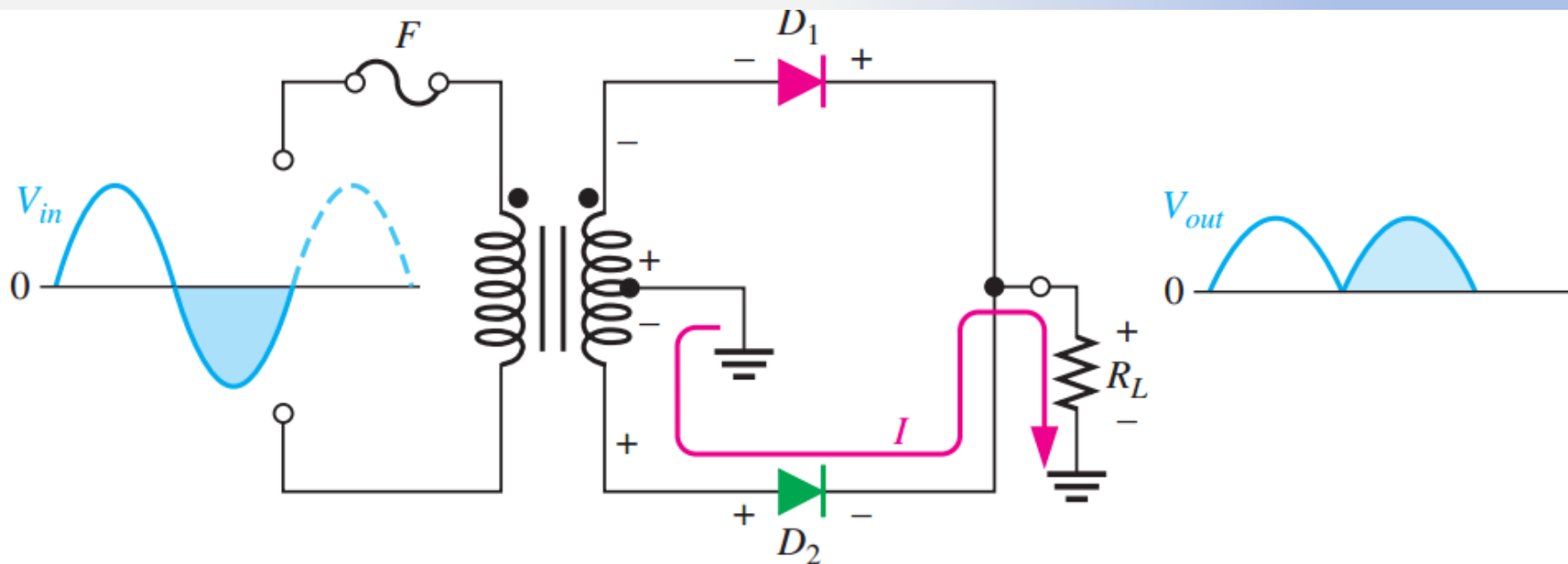
- **Center-Tapped Full-Wave Rectifier: Operation**
- During positive half-cycles:
- **D1** is forward-biased and **D2** is reverse-biased



Applications

FULL-WAVE RECTIFIERS

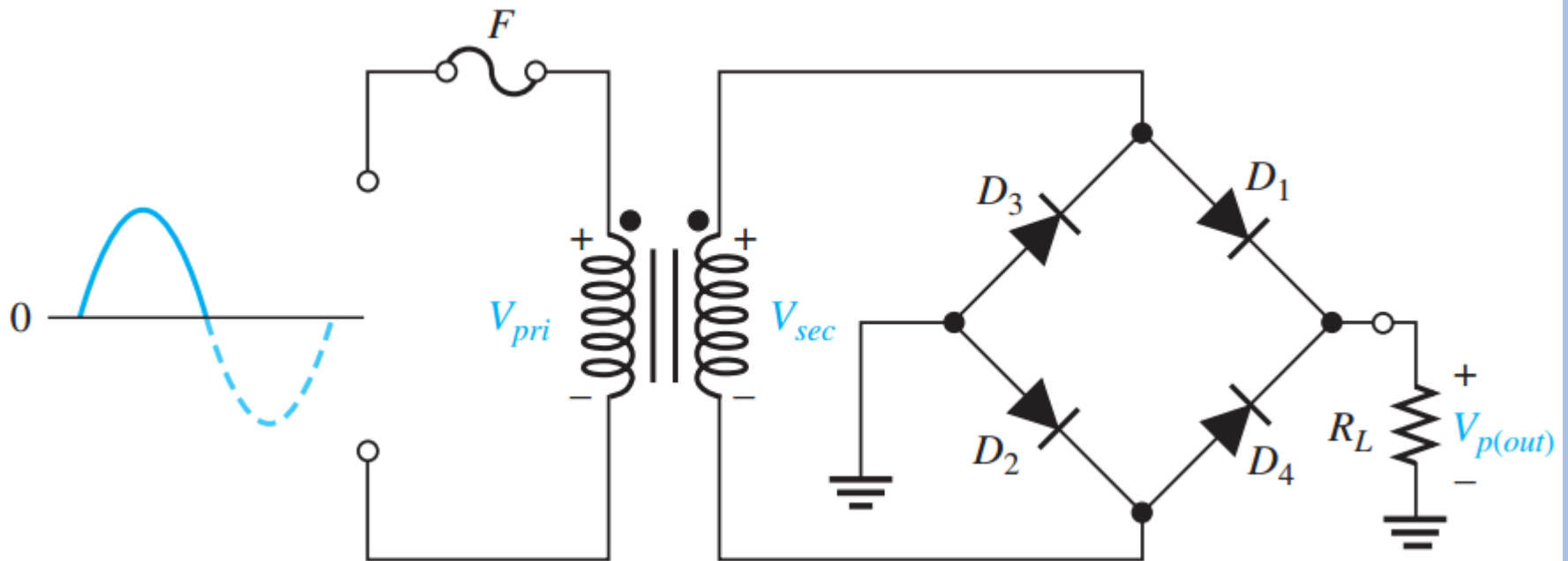
- **Center-Tapped Full-Wave Rectifier: Operation**
- During positive half-cycles:
- **D2** is forward-biased and **D1** is reverse-biased



Applications

FULL-WAVE RECTIFIERS

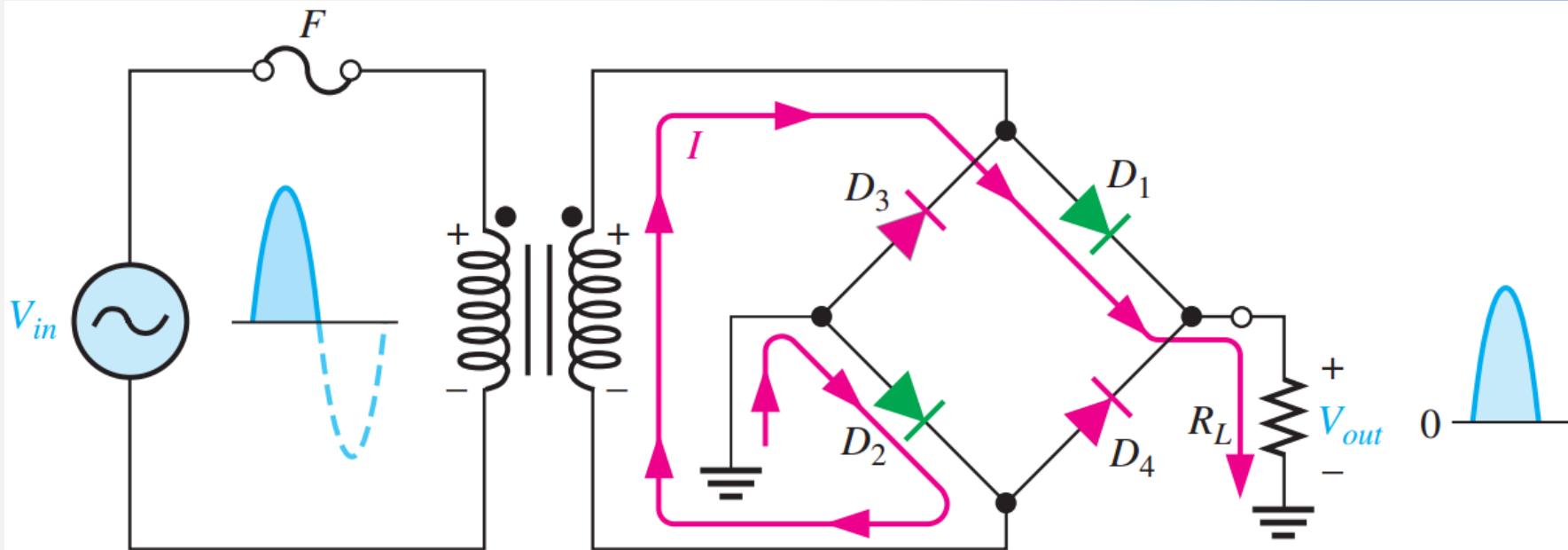
- Bridge Full-Wave Rectifier



Applications

FULL-WAVE RECTIFIERS

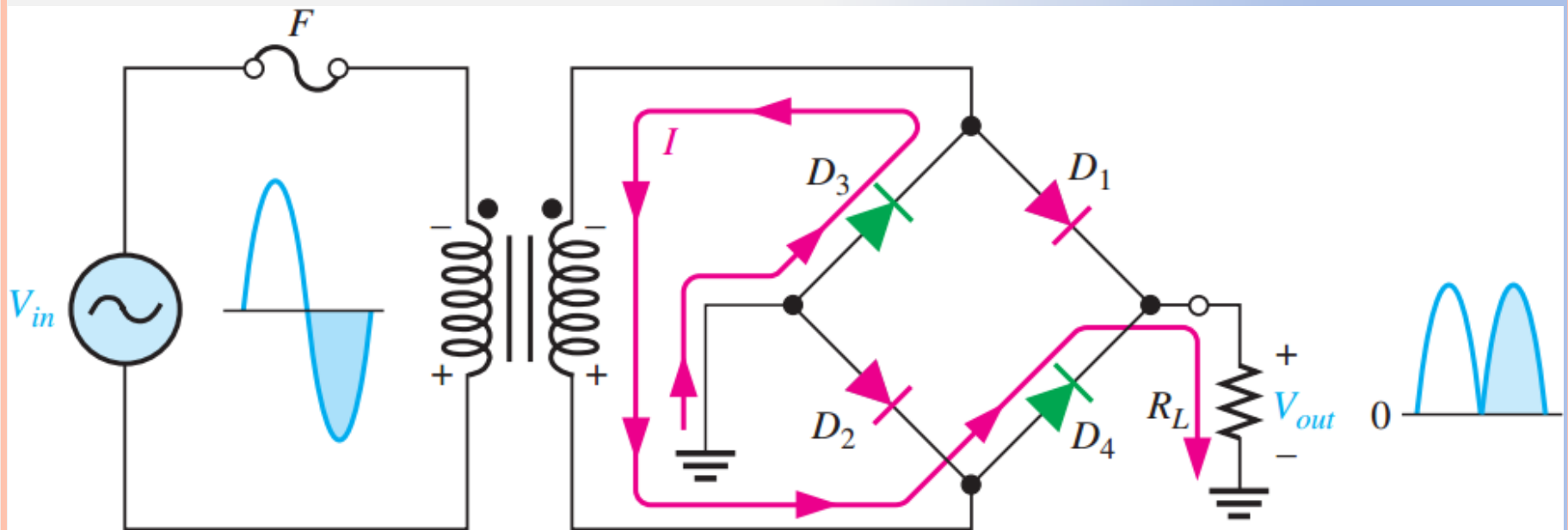
- **Bridge Full-Wave Rectifier: Operation**
- During the positive half-cycle of the input, D_1 and D_2 are forward-biased and conduct current.
- D_3 and D_4 are reverse-biased.



Applications

FULL-WAVE RECTIFIERS

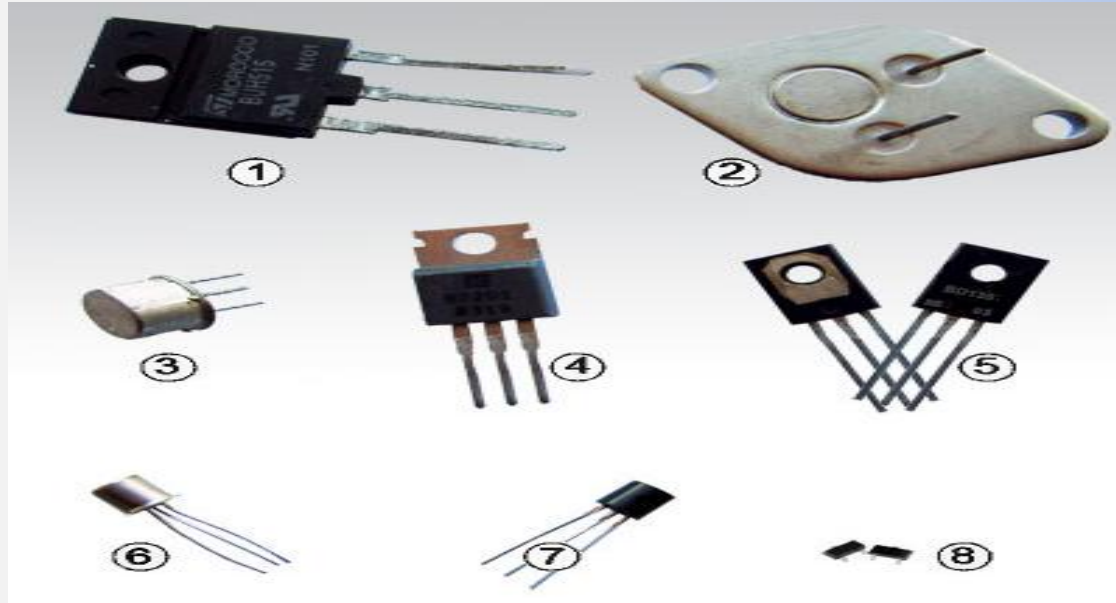
- **Bridge Full-Wave Rectifier: Operation**
- During the negative half-cycle of the input, D_3 and D_4 are forward-biased and conduct current.
- D_1 and D_2 are reverse-biased.



ASSIGNMENTS

- Study the operation and areas of applications of the following special diodes and write short notes on:
 - **Zener Diode**
 - **Point Contact Diode**
 - **Schottky Diode**
 - **Varactor (Vari cap)**
 - **Backward Diode**
 - **Tunnel Diode**
 - **PIN Diode (Positive Intrinsic Diode)**
 - **LED Diode (Light Emitting Diode)**
 - **Laser Diode**
 - **Photodiode**
 - **Clippers and Clampers**

NEXT SESSION ...TRANSISTORS (BJT)



Thank you for listening